

Lecture 2 - SC

Introduction to applied superconductivity

Leonardo CIVALE

Plan of the class

1. Vortex physics

- Review of the main concepts and results of the Ginzburg-Landau theory
- Single vortex – lower critical field H_{c1}
- Vortex-vortex interactions – Abrikosov lattice – upper critical field H_{c2}
- Magnetization curves
- Interaction of vortices and electric currents – vortex motion – flux flow – dissipation

2. Vortex pinning

- Origin and importance of vortex pinning - Critical current density
- Classification of pinning centers
- Vortex elasticity – interactions of many vortices & defects – pinning landscape
- Methods to measure J_c

3. Materials for superconducting wires (LTS & HTS)

4. Pinning enhancement in ReBCO

- Particle irradiation
- Second phases additions

5. CC for fusion

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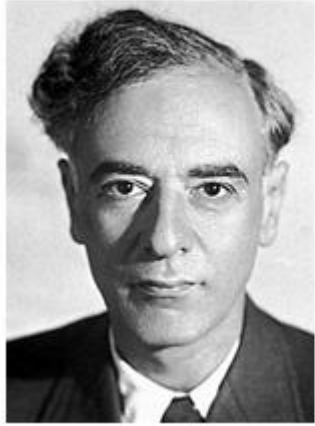
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- Particle irradiation
- Second phases additions

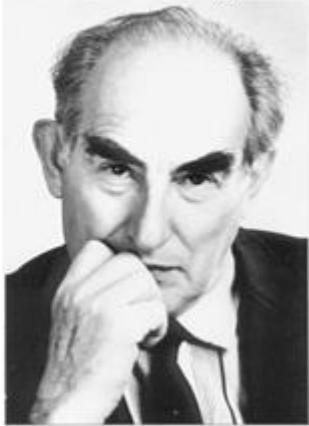
5. CC for fusion

Main concepts of Ginzburg Landau theory (1950)

Landau



Ginzburg



$$f_s = f_{n0} + \alpha|\psi|^2 + \frac{\beta}{2}|\psi|^4 + \frac{1}{2m^*} \left| \left(\frac{\hbar}{i} \nabla - \frac{e^*}{c} \mathbf{A} \right) \psi \right|^2 + \frac{h^2}{8\pi}$$

Basic ideas:

- The order parameter is $\psi=0$ for $T>T_c$ and $\psi \neq 0$ for $T<T_c$
- The transition is second order, ψ increases continuously from T_c
- $|\psi|^2 \sim n_s(T)$ is the T-dependent density of Cooper pairs

Two characteristic lengths determine all the electrodynamic properties of an isotropic superconductor

Penetration
depth

$$\lambda_{GL}(T) = \left(\frac{m}{\mu_0 |\psi|^2 e^2} \right)^{1/2}$$

$|\psi|^2 \sim n_s(T)$ grows from 0 at T_c to $\sim n/2$ at $T \ll T_c$
 n = density of conduction electrons in the normal state

Coherence
length

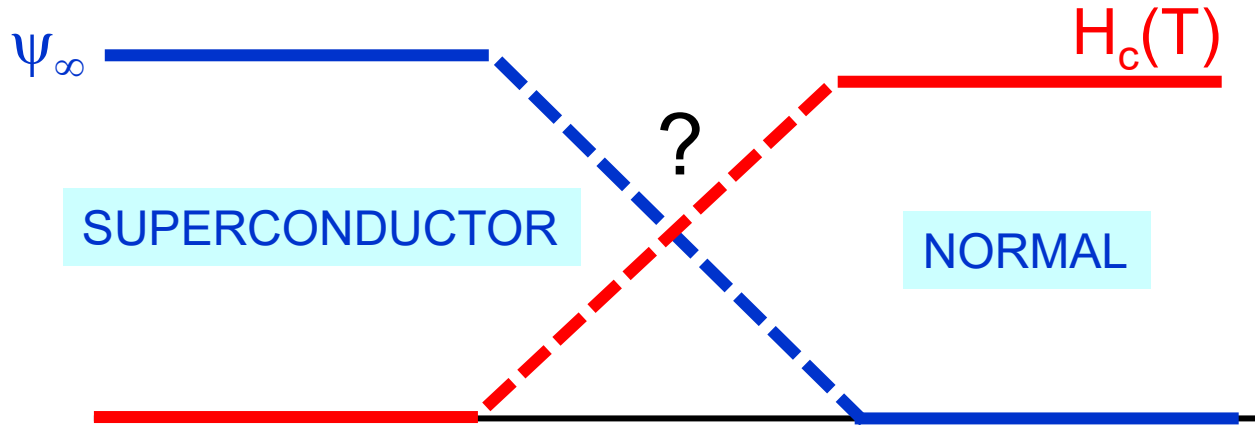
$$\xi_{GL}(T) \sim \frac{\xi_0}{\sqrt{1 - T/T_c}}$$

Minimum scale of spatial variations of $|\psi|^2$

GL parameter	$\kappa = \frac{\lambda}{\xi}$	independent of T
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Domain wall energy

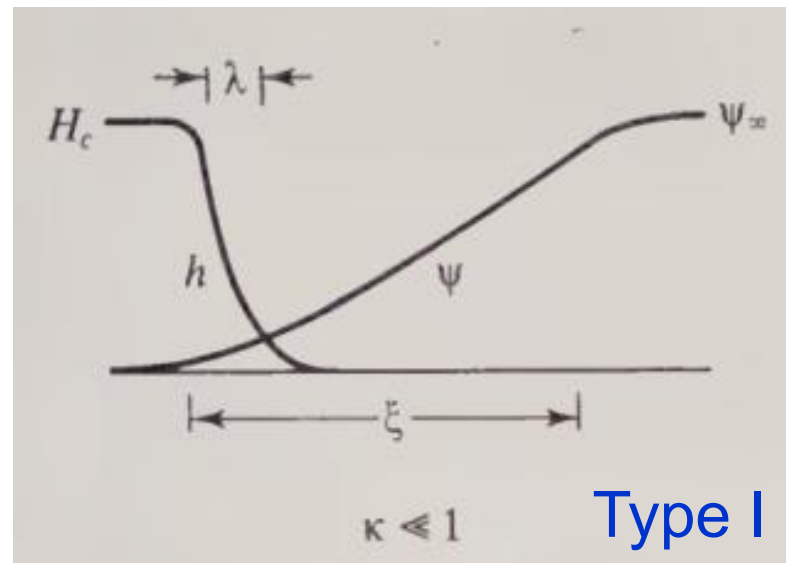
Macroscopic sample at T-H such that the normal & superconducting states have the same free energy



There will be a “domain wall”

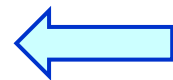
Wall energy (difference with respect to all normal or all superconducting) $\equiv \frac{H_c^2}{8\pi} \delta$

$$\delta = \int_{-\infty}^{\infty} \left[\underbrace{\left(1 - \frac{h}{H_c}\right)^2}_{\text{Range } \sim \lambda} - \underbrace{\left(\frac{\psi}{\psi_\infty}\right)^4}_{\text{Range } \sim \xi} \right] dx$$



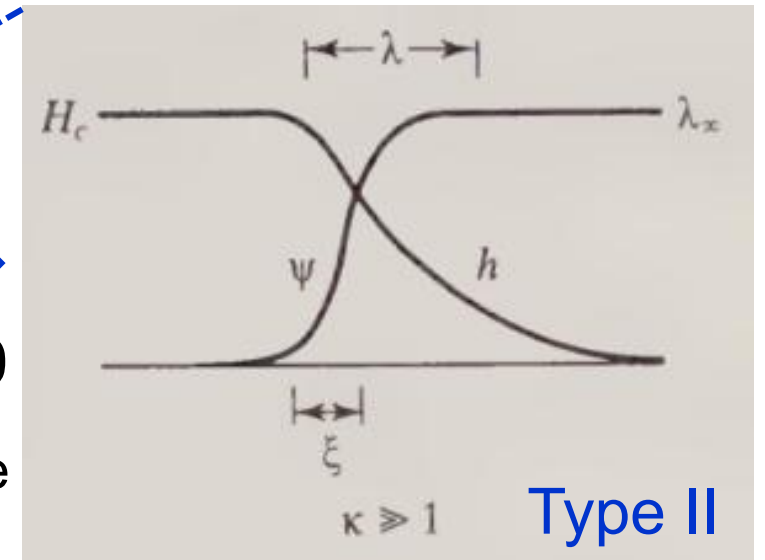
$$\kappa < \frac{1}{\sqrt{2}} \Rightarrow \delta > 0$$

domain walls cost energy
non favorable



$$\kappa > \frac{1}{\sqrt{2}} \Rightarrow \delta < 0$$

domain walls lower the
energy – favorable!!

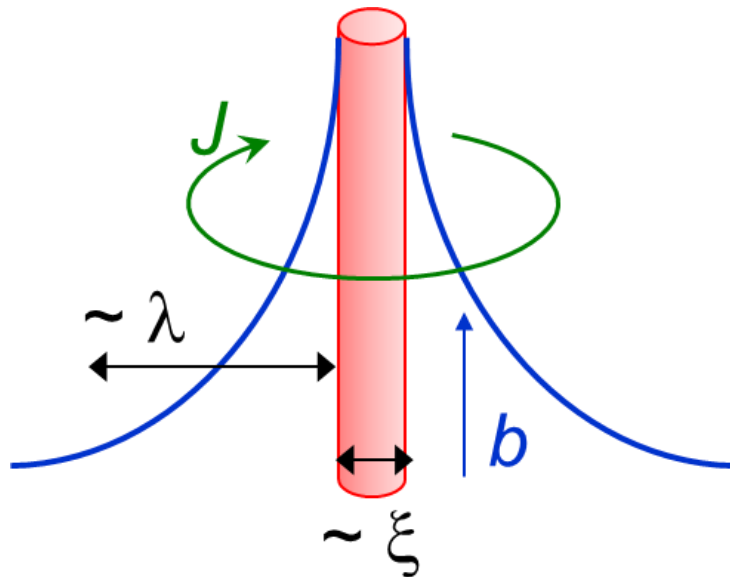


Type II superconductors - vortices

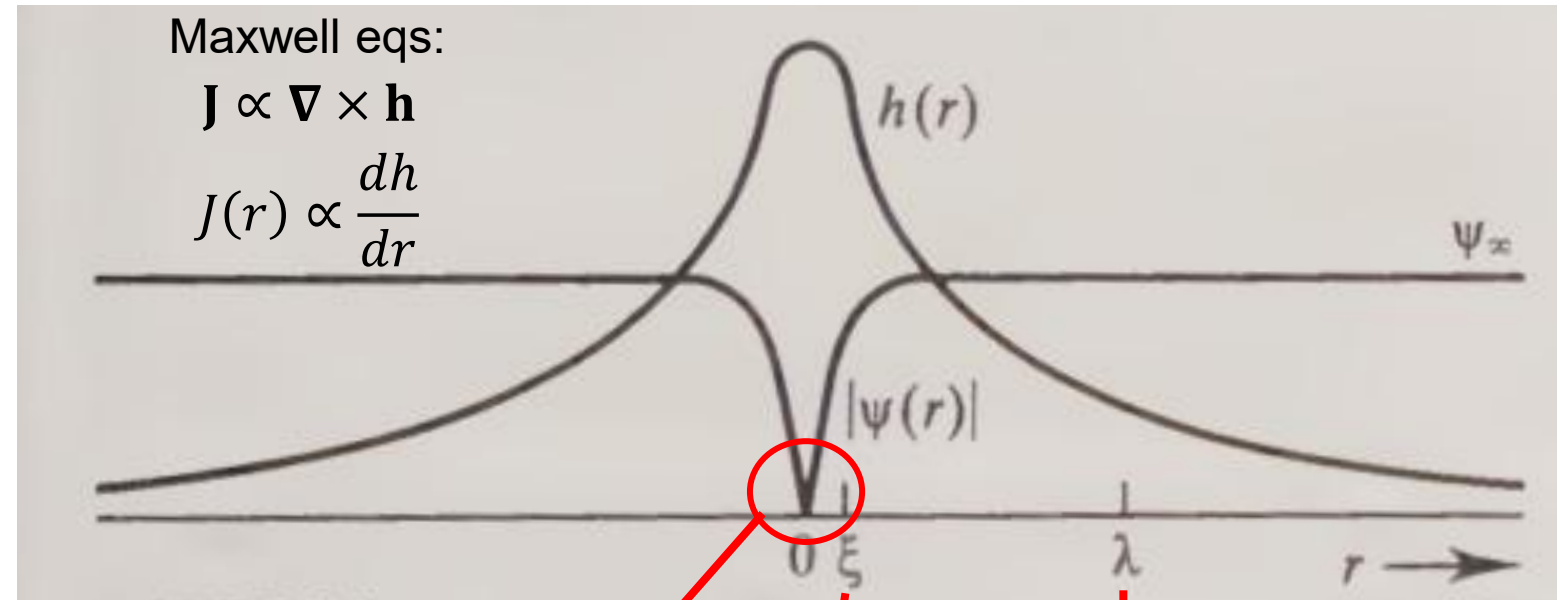


Alexei Abrikosov

- If domain walls are energetically favorable, the more the better. Where does it stop?
- Many walls will form. How will they interact?
- Would this enable superconductivity to be retained up to higher magnetic fields?
- Diverse configurations are possible (e.g., laminar)
- Alexei Abrikosov showed that the most energetically favorable solution is the vortex lattice (paper **published** in 1957)



A superconducting vortex is a “tube” rotating supercurrents that generate axial field, with a **core** at the center, where superconductivity is suppressed



ψ MUST be zero at the axis

Scale of variations of ψ

Scale of variations of h and J

Isolated Vortex - H_{c1}

We seek for a minimum in GL free energy (solution to the GL differential equations).

In cylindrical coordinates (r, θ, z) , we focus on configurations with axial symmetry: $|\psi|^2$, $\mathbf{h} = h(r)\mathbf{e}_z$ and $\mathbf{J} = J(r)\mathbf{e}_\theta$ are only function of r .

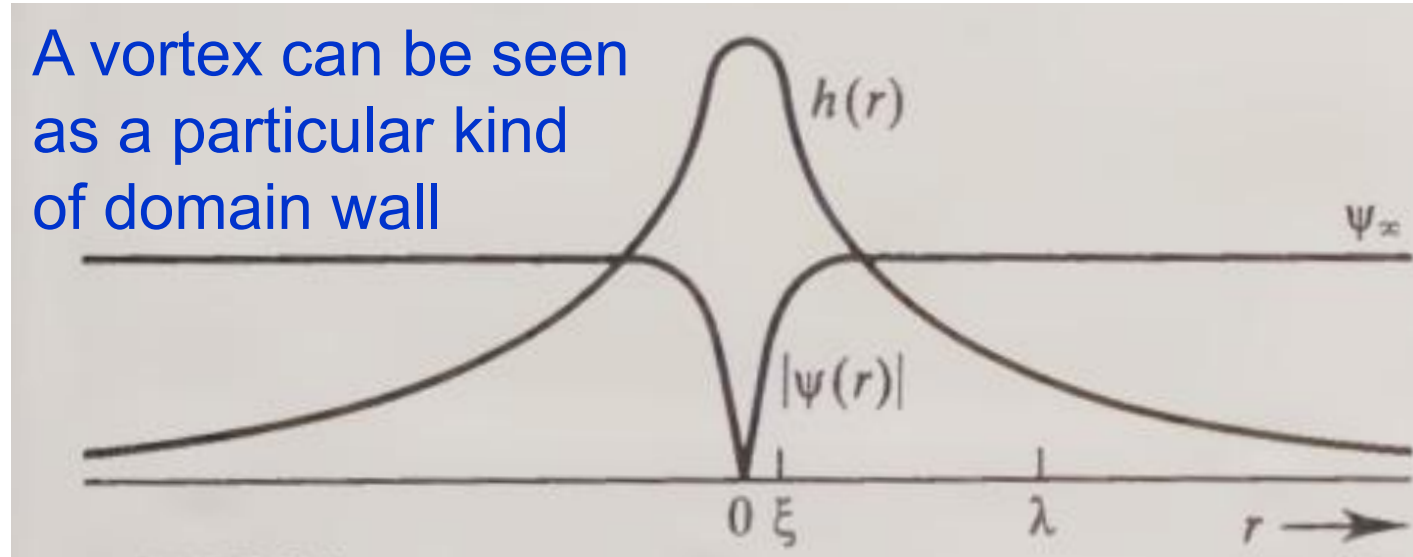
The magnetic flux of a vortex is quantized

$$\Phi = \int_0^\infty h(r) dr = \frac{hc}{2e} \equiv \Phi_0$$

Flux quantum:

$$\begin{aligned}\Phi_0 &= 2.07 \times 10^{-7} \text{ G}\cdot\text{cm}^2 \text{ (CGS)} \\ &= 2.07 \times 10^{-15} \text{ weber (MKS)}\end{aligned}$$

A vortex can be seen as a particular kind of domain wall



- Nucleating a vortex lowers the free energy as compared to the Meissner state (reduces magnetic pressure), by
$$-\frac{H}{4\pi} \int_0^\infty h(r) dr = -\frac{H\Phi_0}{4\pi} \text{ per unit length}$$
- But, a vortex has energy: magnetic $\propto h^2$, kinetic of the currents $\propto J^2$, condensation energy lost inside the core. The energy *per unit length* is ε_1 (**line energy** or **line tension**)

The penetration of a vortex becomes favorable at H_{c1} where the Gibbs free energy with zero (=Meissner) and one vortex are equal (both terms compensate)

$$H_{c1} = \frac{4\pi\varepsilon_1}{\Phi_0} \quad \text{lower critical field}$$

Isolated Vortex - high κ approximation

For $\kappa \gg 1$, an analytical solution is possible. Outside the core $|\psi|^2 = \text{constant}$, $\Rightarrow$ we can use the London eqs. The core radius is very small, we can replace it by a delta function.

$$\nabla^2 \mathbf{h} - \frac{\mathbf{h}}{\lambda^2} = \text{X} \quad \nabla^2 \mathbf{h} - \frac{\mathbf{h}}{\lambda^2} = -\frac{\Phi_0}{\lambda^2} \delta_2(\mathbf{r}) \hat{\mathbf{z}} \quad \rightarrow \quad \frac{d^2 h}{dr^2} - \frac{h}{\lambda^2} = -\frac{\Phi_0}{\lambda^2} \delta_2(r)$$

Exact solution: $h(r) = \frac{\Phi_0}{2\pi\lambda^2} K_0\left(\frac{r}{\lambda}\right)$ [K_0 is a zeroth-order Hankel function of imaginary argument]

$h > 0$ everywhere

Limiting expressions: $h(r) \sim \frac{\Phi_0}{2\pi\lambda^2} \times \begin{cases} e^{-r/\lambda} & \text{for } r \gg \lambda \\ \ln(r/\lambda) & \text{for } \xi < r < \lambda \end{cases} \Rightarrow \begin{array}{l} \text{short range interactions} \\ h(r) \sim \frac{\Phi_0}{2\pi\lambda^2} \ln(\kappa) \text{ for } r \leq \xi \end{array}$

$$J(r) \propto \frac{dh}{dr} \propto \begin{cases} e^{-r/\lambda} & \text{for } r \gg \lambda \\ \frac{1}{r} & \text{for } \xi < r < \lambda \end{cases}$$

At the core radius:

$$J(\sqrt{2}\xi) \sim \frac{\Phi_0}{2\sqrt{2}\pi\lambda^2\xi} \sim \frac{H_c}{\lambda} \sim J_d \text{ depairing!}$$

$$\varepsilon_1 = \left(\frac{\Phi_0}{4\pi\lambda}\right)^2 \ln(\kappa) \quad \text{Vortices penetrate below } H_c$$

$$H_{c1} = \frac{\Phi_0}{4\pi\lambda^2} \ln(\kappa) = \frac{H_c}{\sqrt{2}\kappa} \ln(\kappa) < H_c$$

Interaction between two vortices

We will remain in the high κ approximation $\Rightarrow$ London equations $\Rightarrow$ linear $\Rightarrow$ we can use superposition:

The total magnetic field $\mathbf{h}(\mathbf{r})$ and current $\mathbf{J}(\mathbf{r})$ produced by N vortices at locations $\mathbf{r}_1, \dots, \mathbf{r}_N$ are the superposition (sum) of the contributions of the individual vortices.

Basic result: vortices repel each other

We can show this by evaluating the energy:

2 vortices, at $\mathbf{r}_1$ and $\mathbf{r}_2$: $\mathbf{h}(\mathbf{r}) = \mathbf{h}_1(\mathbf{r}) + \mathbf{h}_2(\mathbf{r}) = [h(|\mathbf{r}-\mathbf{r}_1|) + h(|\mathbf{r}-\mathbf{r}_2|)]\mathbf{e}_z$
 $\mathbf{J}(\mathbf{r}) = \mathbf{J}_1(\mathbf{r}) + \mathbf{J}_2(\mathbf{r}) = \dots$ (careful with the vector components here)

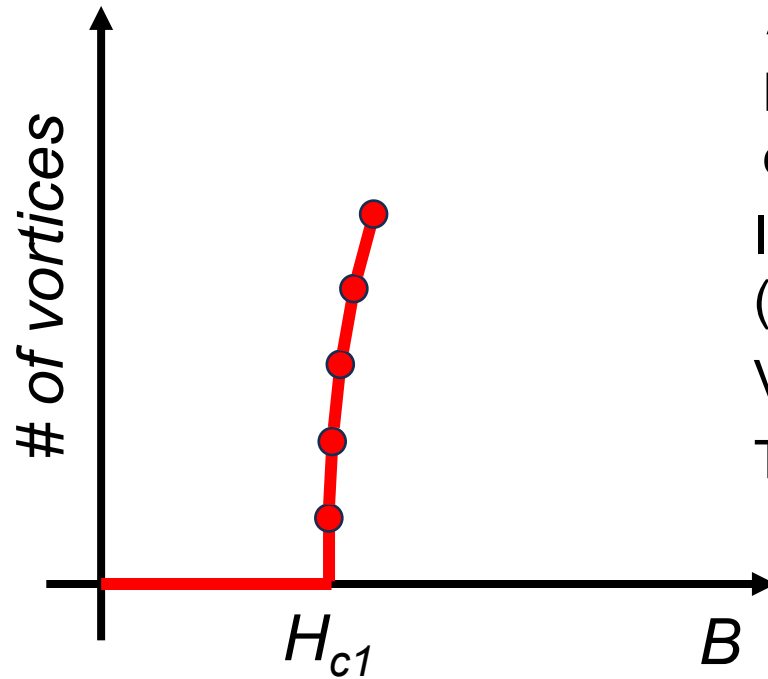
Magnetic energy

$$\propto \int h^2 d\mathbf{r} \sim \int [h(|\mathbf{r}-\mathbf{r}_1|) + h(|\mathbf{r}-\mathbf{r}_2|)]^2 d\mathbf{r} \sim \underbrace{2 \int [h_1(\mathbf{r})]^2 d\mathbf{r}}_{2\varepsilon_1} + \underbrace{2 \int [h(|\mathbf{r}-\mathbf{r}_1|) * h(|\mathbf{r}-\mathbf{r}_2|)] d\mathbf{r}}_{\text{Magnetic energy} > 0 \Rightarrow \text{the farther the lower}}$$

The same argument explains why there are typically not $2\Phi_0$ vortices

An alternative approach is to calculate the repulsive force directly.
We will do that later, when discussing vortex motion and dissipation.

The Abrikosov vortex lattice



At H_{c1} the first vortex penetrates. Then what?

Each new vortex will further reduce the free energy $\Rightarrow$ they will keep entering...until the repulsion from previous vortices stop them.

In contrast to long-range repulsion between electrically charged particles ($\propto r^{-2}$), inter-vortex repulsion is short range (exponential).

Vortices will penetrate in avalanche until they are a few λ apart.

They will form a periodic triangular lattice, the **Abrikosov lattice**.

Lattice parameter decreases with B

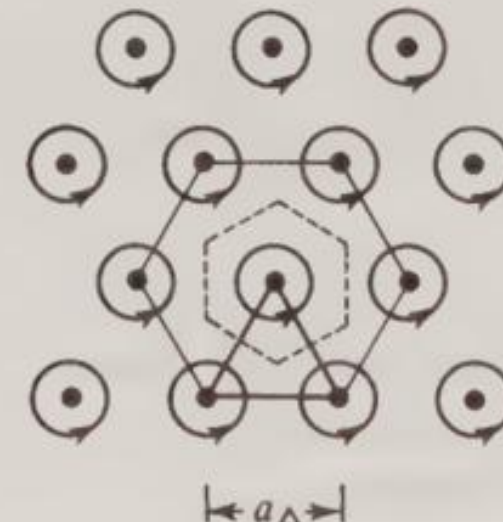
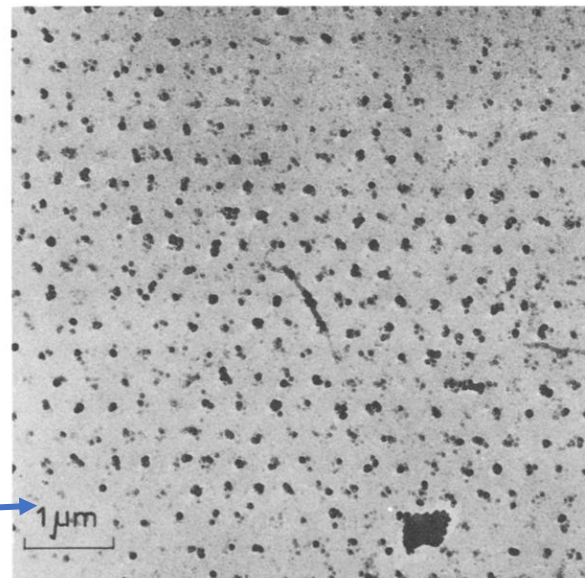
$$a_{\Delta} = \left(\frac{4}{3}\right)^{1/4} \left(\frac{\Phi_0}{B}\right)^{1/2} = 1.075 \left(\frac{\Phi_0}{B}\right)^{1/2}$$

Historical curiosity:

Abrikosov originally concluded that the ground state (lowest free energy) was a square lattice.

The energy difference between the two is very small.

The vortex lattice is triangular

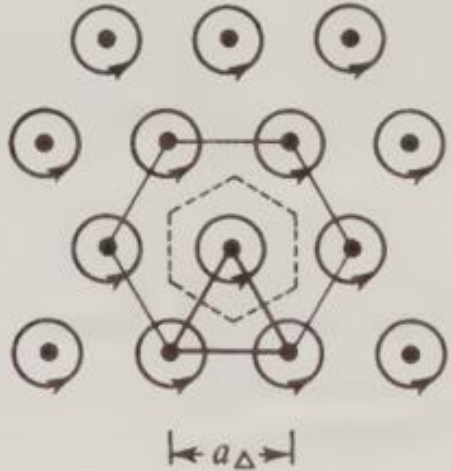


Smaller than, but close to, the applied field (except very near H_{c1})

The upper critical field H_{c2}

$$a_{\Delta} = \left(\frac{4}{3}\right)^{1/4} \left(\frac{\Phi_0}{B}\right)^{1/2} = 1.075 \left(\frac{\Phi_0}{B}\right)^{1/2}$$

$$\sim 2\sqrt{2}\xi$$



At high enough field, vortices get so close that the cores overlap
 $\Rightarrow$ all the volume is occupied by non-superconducting cores $\Rightarrow$
 $\Rightarrow$ no more superconductivity $\Rightarrow$ **Upper critical field H_{c2}**
 (technically, no more vortex solutions of the G.L. equations)

$$\text{Estimate: } H_{c2} \sim \frac{\Phi_0}{8\xi^2}$$

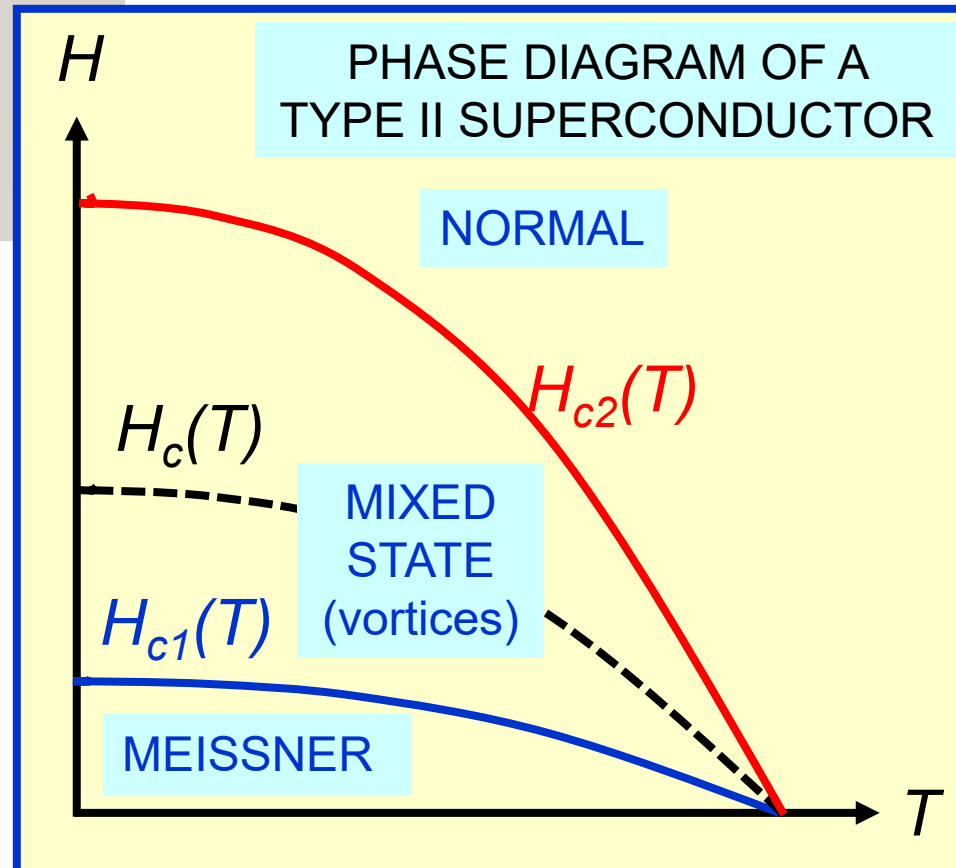
Exact result:

$$H_{c2} = \frac{\Phi_0}{2\pi\xi^2} = \sqrt{2}\kappa H_c > H_c$$

$$H_{c1}H_{c2} = H_c^2 \ln(\kappa)$$

Simple rule (not perfect):

- Elements are type I
- Alloys & Compounds are type II
- HTS are type II



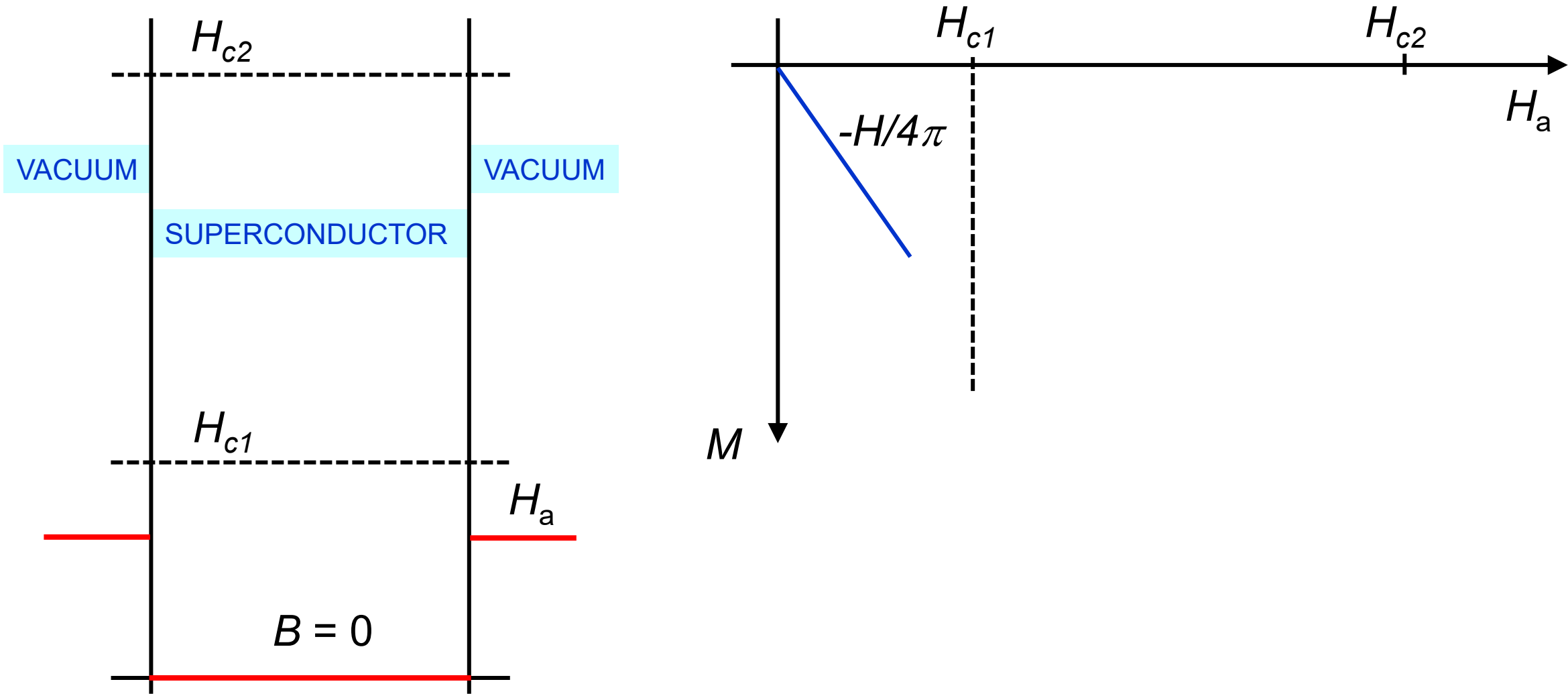
The good news:

Vortices enable superconductivity up to much higher magnetic fields

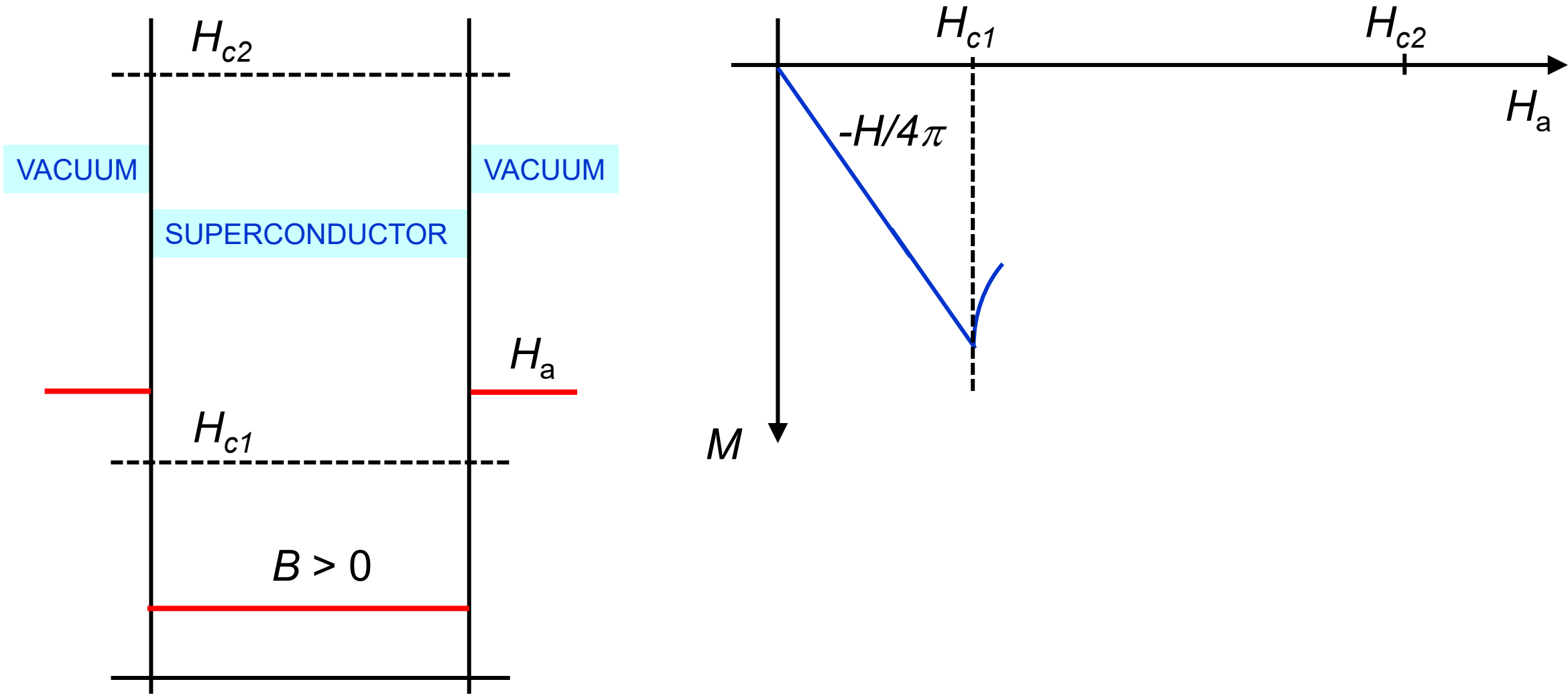
For instance:

In HTS, such as YBCO,
 $H_c \sim 1\text{T}$ and $\kappa \sim 100 \Rightarrow H_{c2} > 100\text{T} !!$

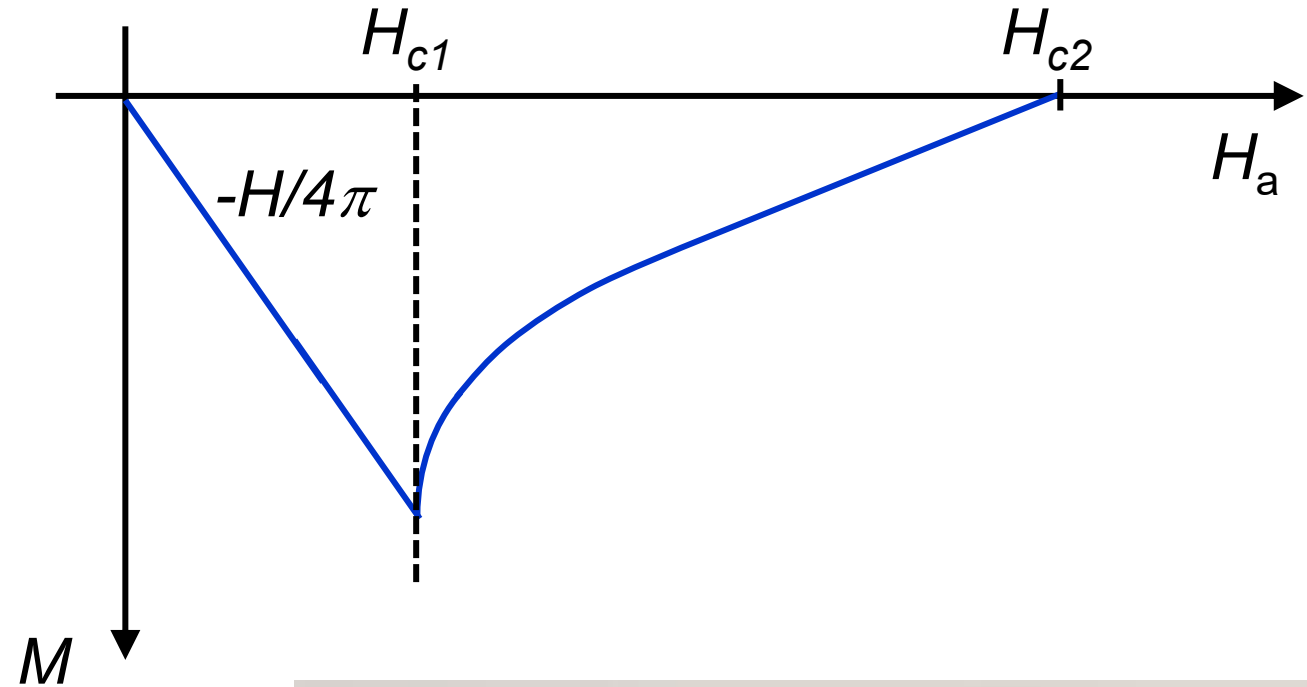
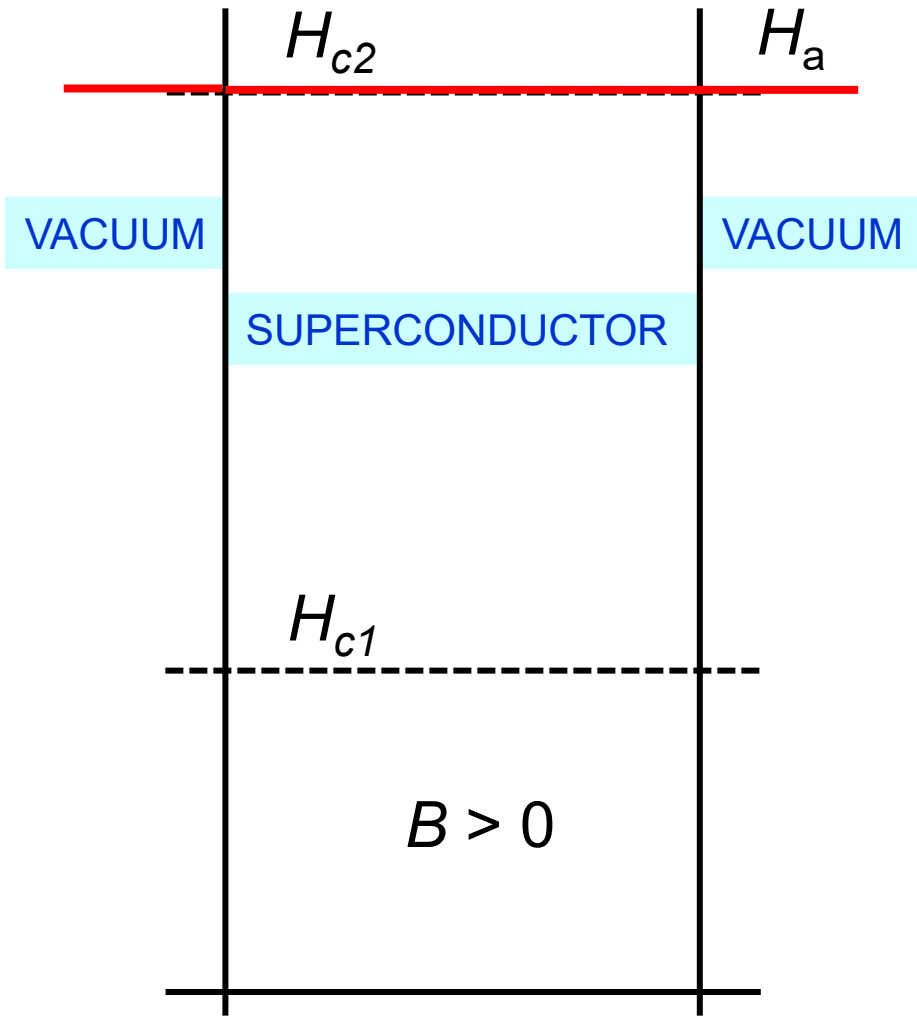
Magnetization curves



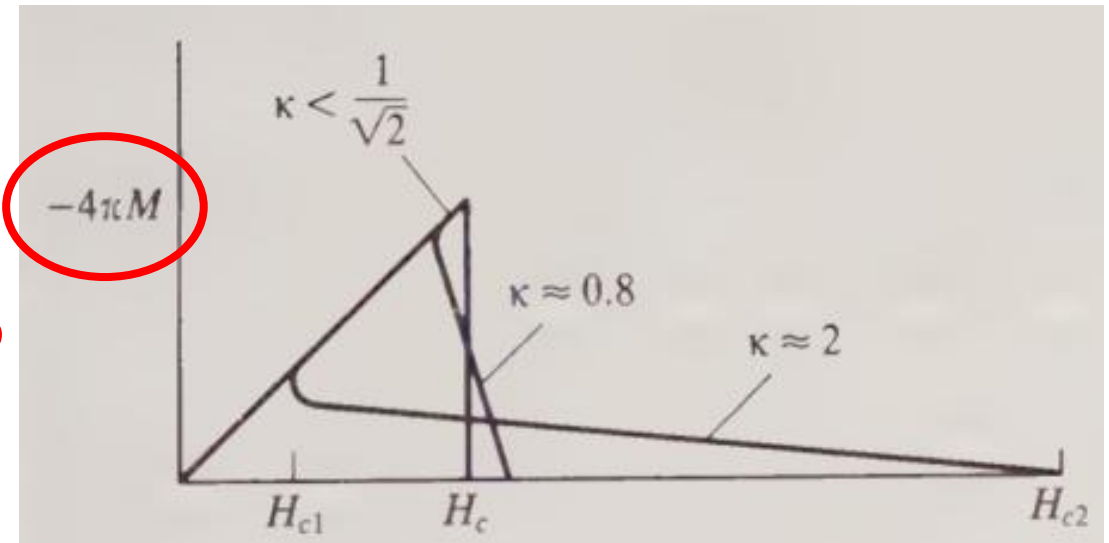
Magnetization curves



Magnetization curves



Sometimes the vertical axis is plot as $-M$ (I do not like it)



Forces that electric currents exert on vortices - flux flow

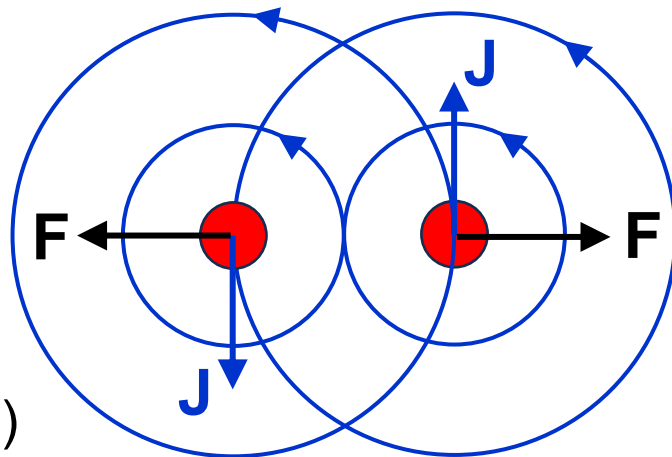
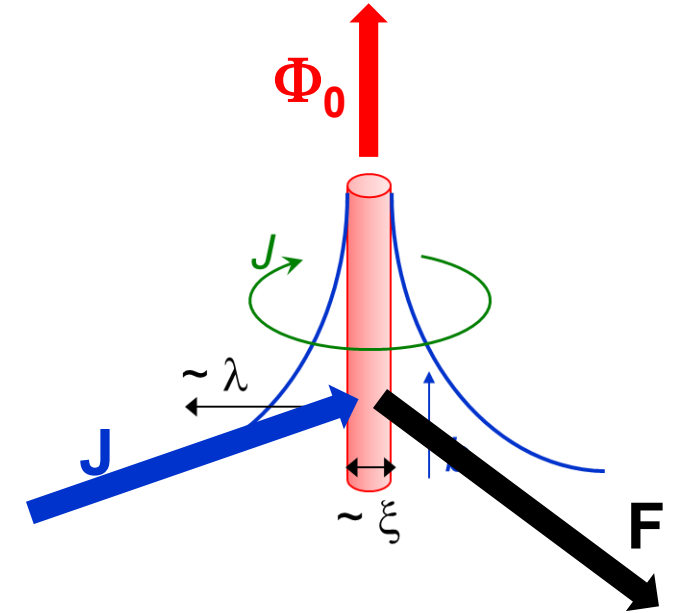
Force on the vortex

$$\mathbf{F} = \mathbf{J} \times \Phi_0$$

Vector of magnitude Φ_0 and direction of the vortex magnetic field

Electric current from any source other than the vortex (transport, Meissner screening, another vortex)

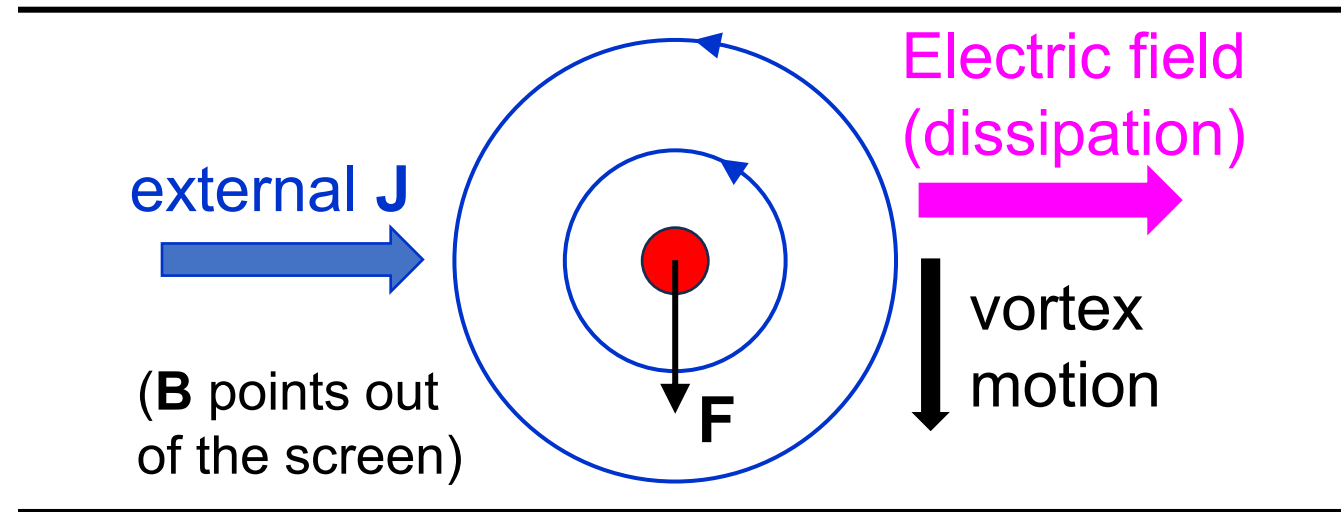
This looks exactly like a Lorentz force $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$
In reality, it is NOT a Lorentz force....
....but everybody calls it a Lorentz force



($\mathbf{B}$ points out of the screen)

Alternative way to show that vortices repel each other

Vortices moving - flux flow



The bad news:

An external applied $\mathbf{J}$ (e.g., in a superconducting magnet), exerts forces on the vortices $\Rightarrow$
 $\Rightarrow$ If vortices move (flux flow regime), dissipation appears due to motion of non-superconducting cores
 $\Rightarrow$ It looks like an ohmic resistance $\Rightarrow R > 0 \Rightarrow$ any benefit of a superconducting wire is lost!!

Fortunately, vortex pinning comes to the rescue!

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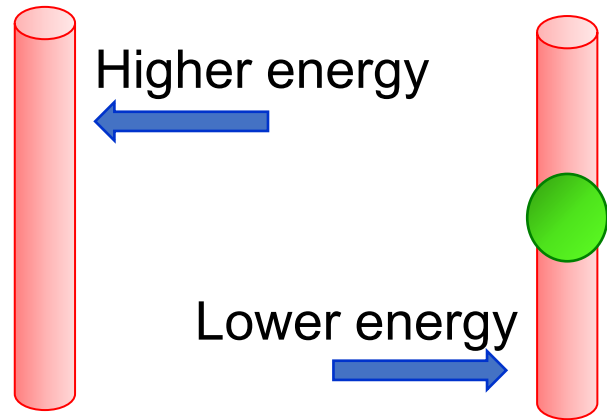
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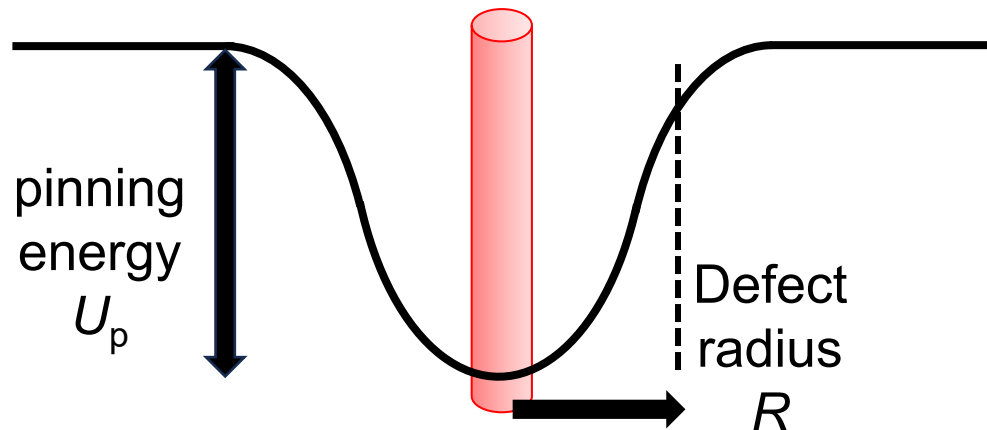
Vortex pinning

Vortex energy: ε_1 = magnetic + kinetic of the currents + condensation energy lost inside the core



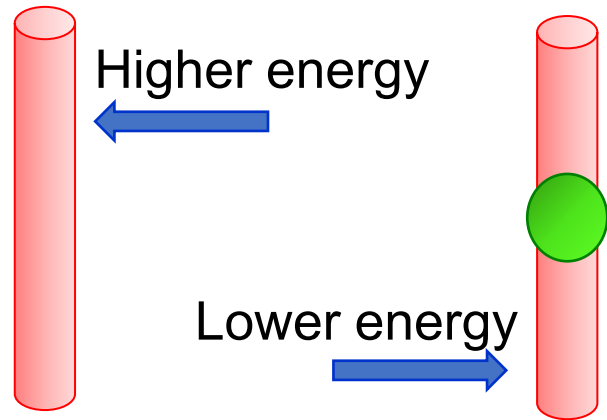
If a vortex sits on a portion of the material that is not superconducting (a defect), this energy cost disappears $\Rightarrow$ vortex costs less energy
The energy reduction is called **pinning energy**

Pinning center:



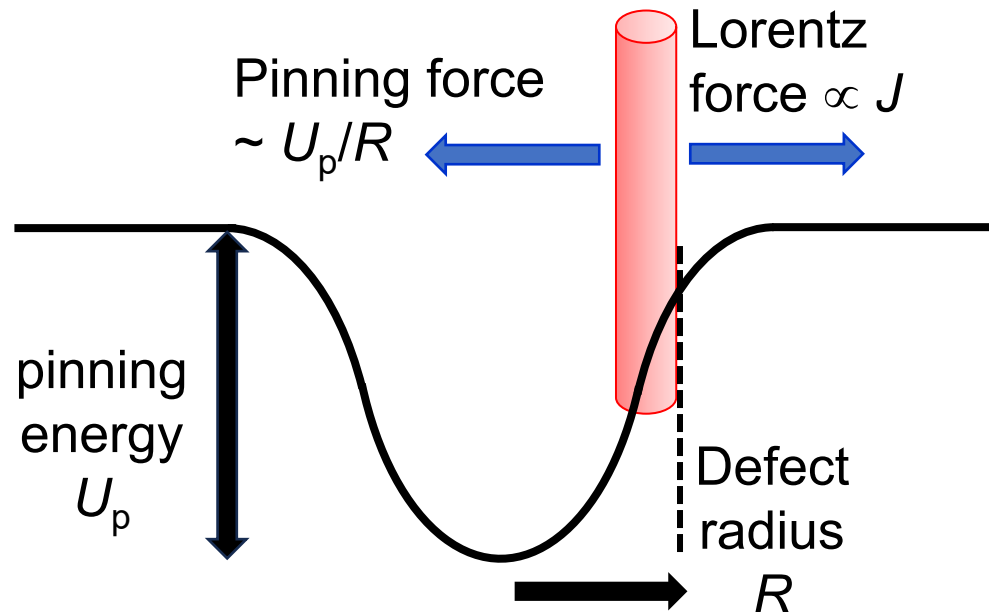
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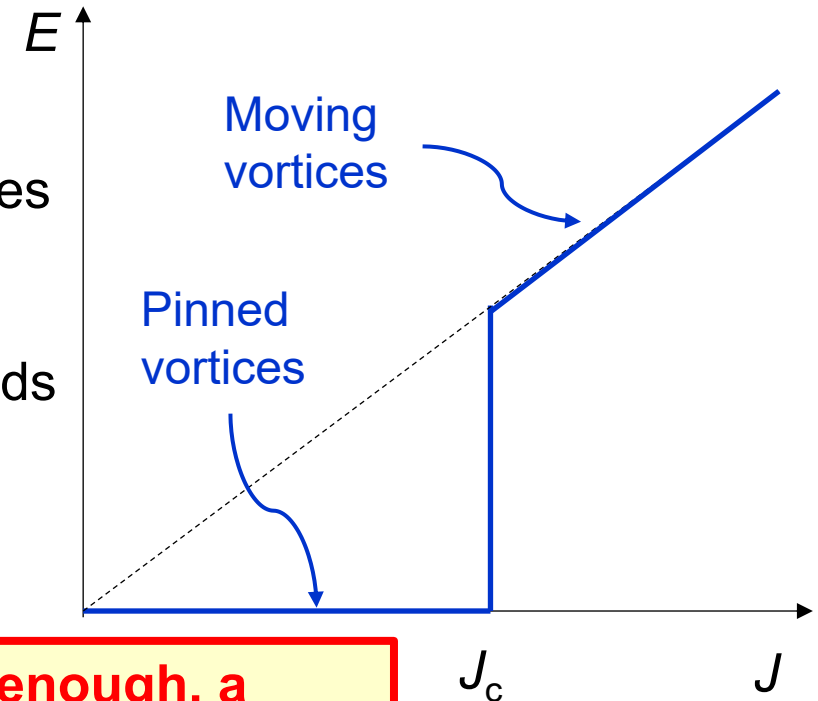


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Effect of an applied current:



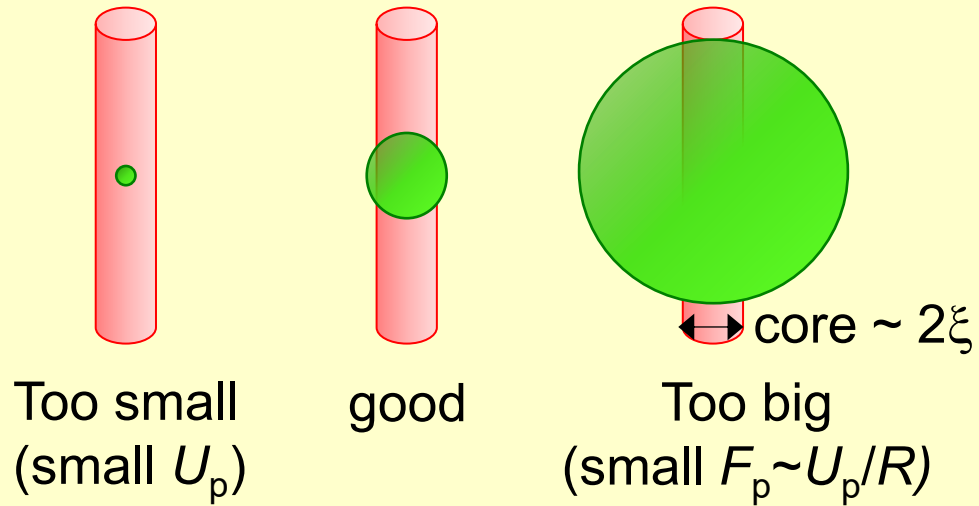
Vortices remain pinned until J reaches the **critical current density J_c** , where Lorentz force exceeds maximum pinning force



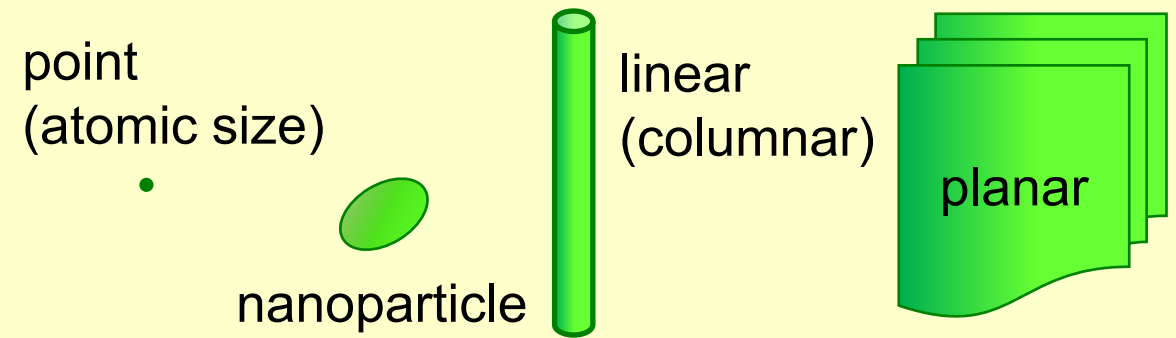
If J_c is not high enough, a superconducting wire is useless

Many types of defects can act as pinning centers, some are better than others...

Size

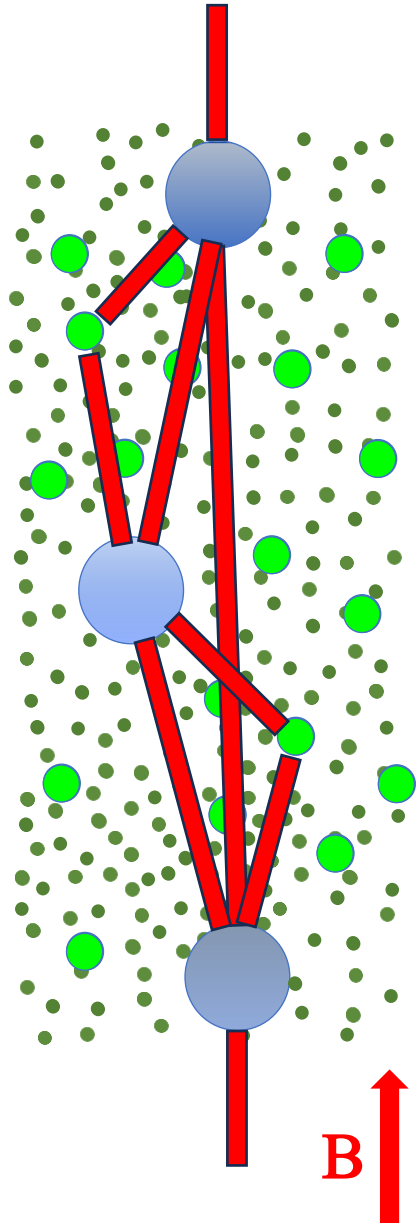


Shape



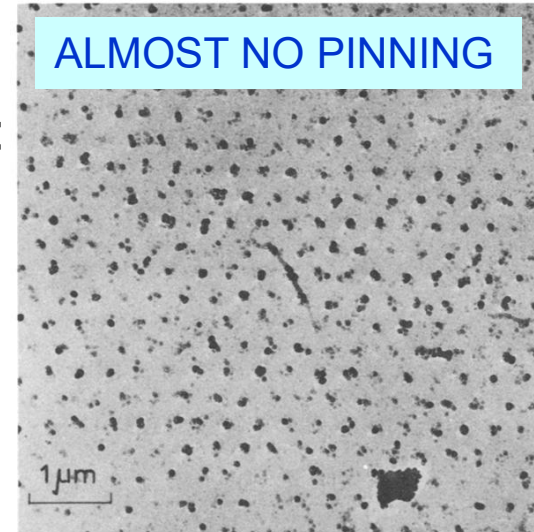
- J_c is an **EXTRINSIC** property $\Rightarrow$ can change by orders of magnitude in the same material
- More defects $\Rightarrow$ more $F_p \Rightarrow$ higher $J_c \Rightarrow$ the more defects the better, correct?
- Yes, to some extent, but **PINNING IS NOT ADDITIVE**
- Complex many-body problem: vortices interacting with many defects and with each other
- There are competing effects: adding new defects could be useless or even detrimental

Vortices are elastic objects - after all, they are just current and field!

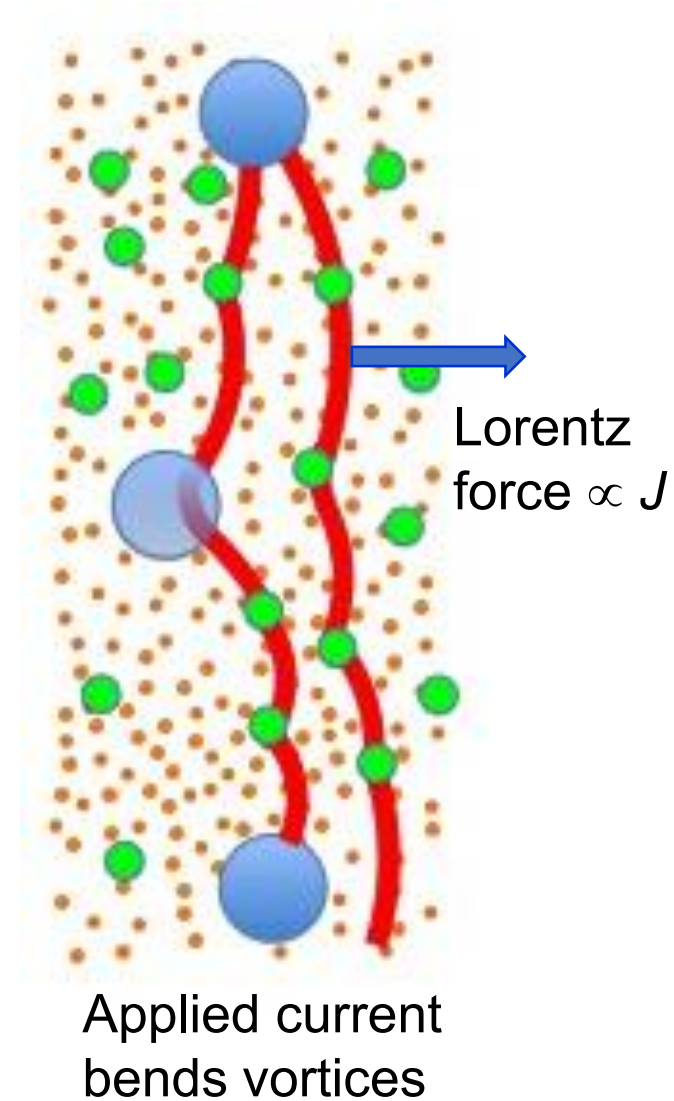
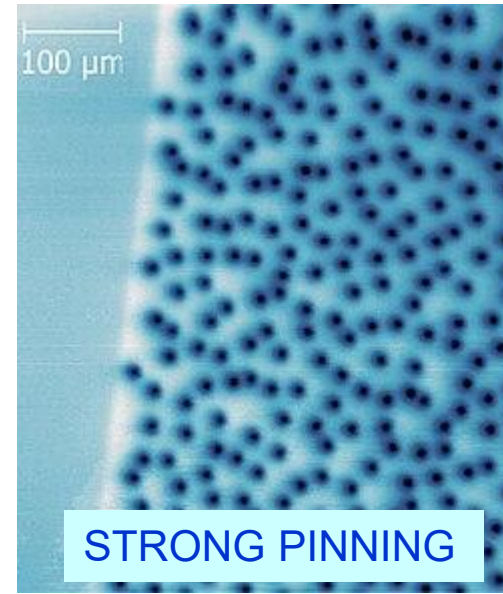


Getting pinned by more defects:

- more gain in pinning energy
- But, tilt with respect to $B \Rightarrow$ cost in magnetic energy
- But, longer length $\Rightarrow$ cost in elastic energy
- What is the best path?
- Difficult optimization problem depends on T and B



the Abrikosov lattice will turn into a disordered array



J_c depends on T, B, and field orientation

CC can operate at very high fields

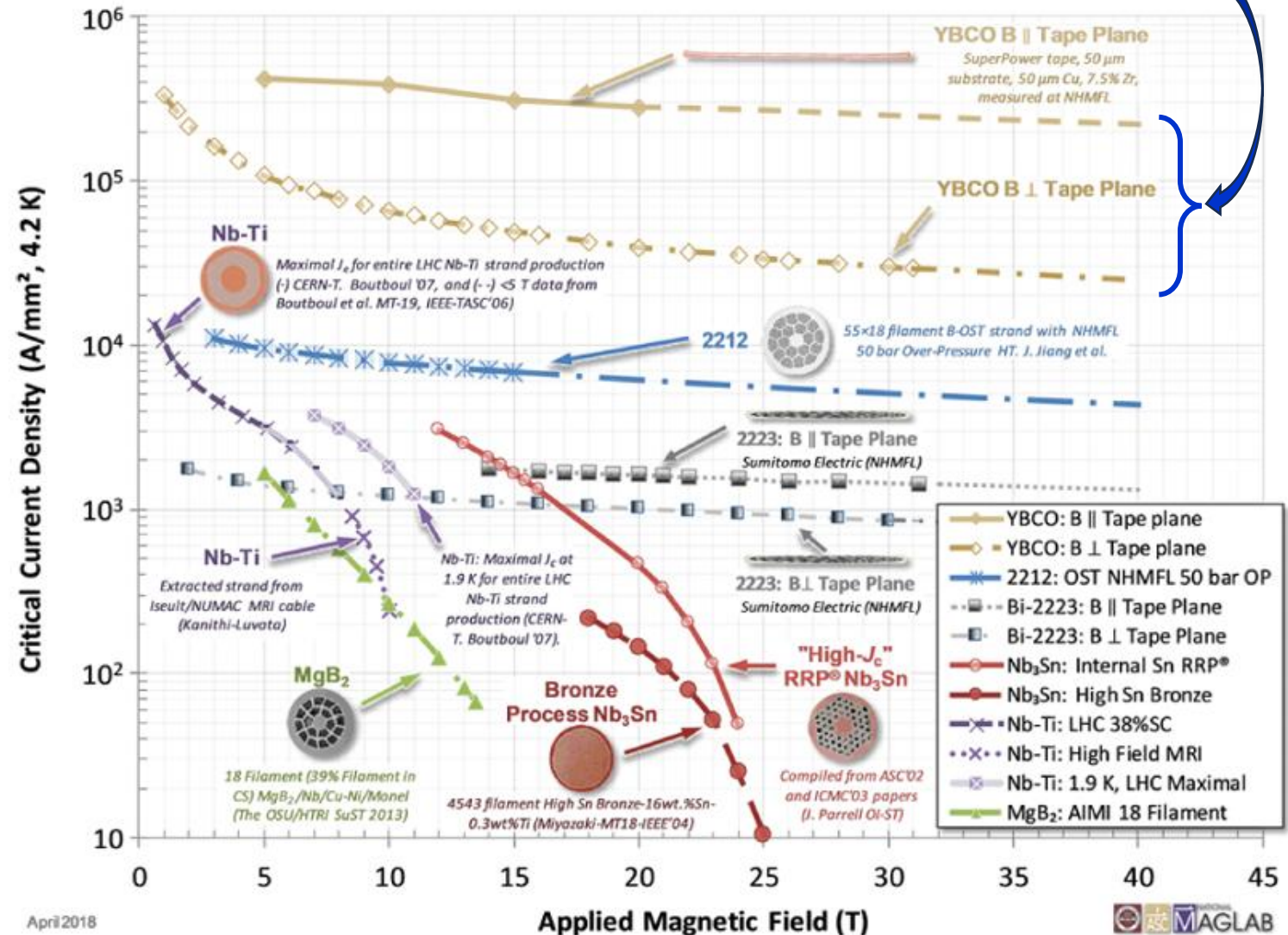
Usually, J_c decreases with T because

- U_p (related to ε_1) decreases with T
- The core radius ($\propto \xi$) increases with T
- Thermal fluctuations of vortices

Usually, J_c decreases with B because

- There may not be enough defects
- The best defects are already taken!
- Thermal fluctuations of vortices

If the pinning is directional (linear or planar defects) or the material is anisotropic, J_c depends on the field orientation (more later)



April 2018

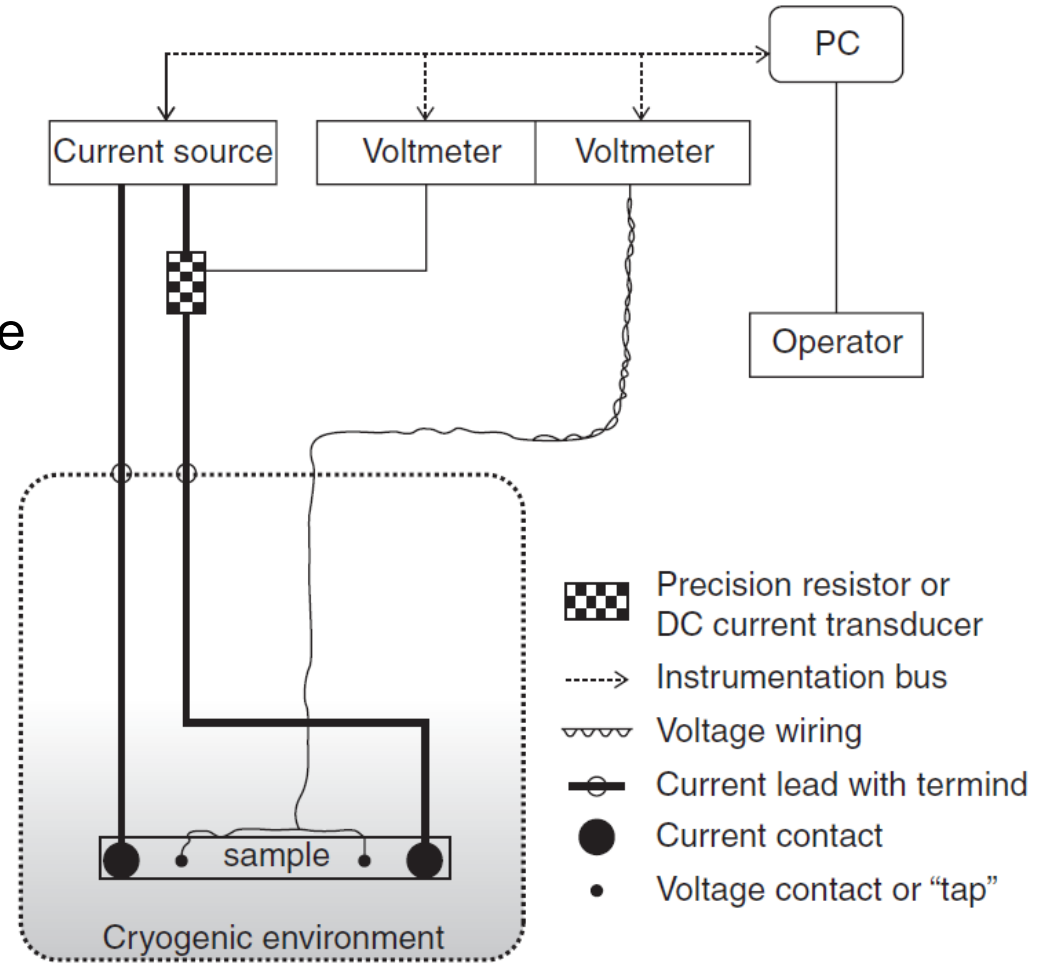
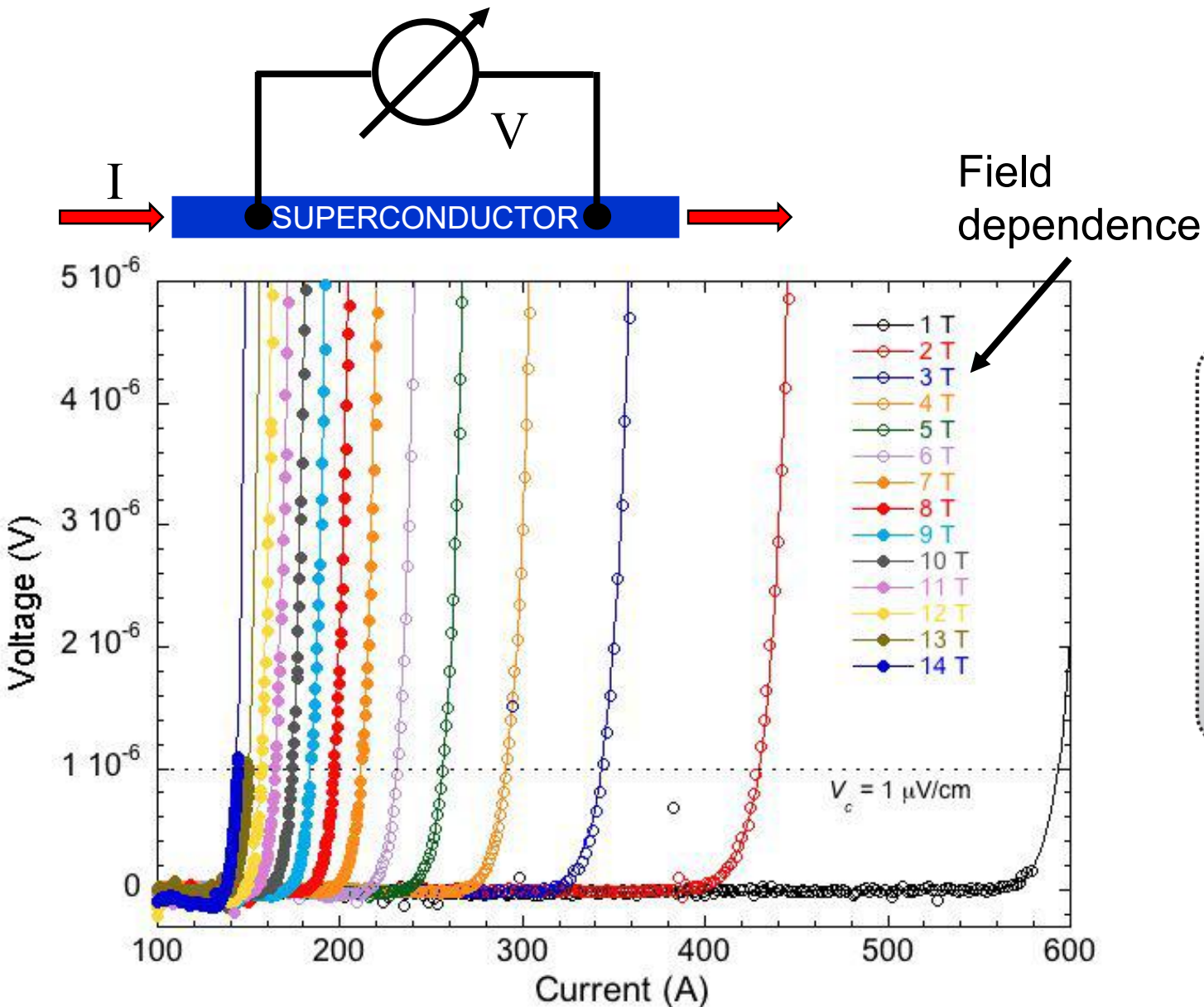
Methods to increase J_c

Finding methods to increase J_c (both in LTS and HTS) has been the main focus of applied superconductivity research for several decades
(In particular, it has been the main focus of my career)

- The idea is to create an effective vortex pinning landscape by nanoengineering the material defects.
- It is common to distinguish between natural defects (which occur during fabrication) and deliberately introduced Artificial Pinning Centers (APC), although that classification is a bit arbitrary.
- The two main methods to create APC are
 - Radiation damage (material defects created by particle irradiation)
 - Chemical incorporation of non-superconducting second phases

Methods to measure J_c

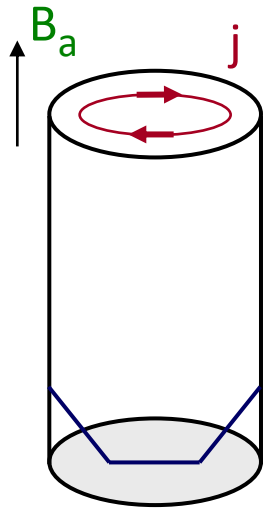
1. Transport (four probe)



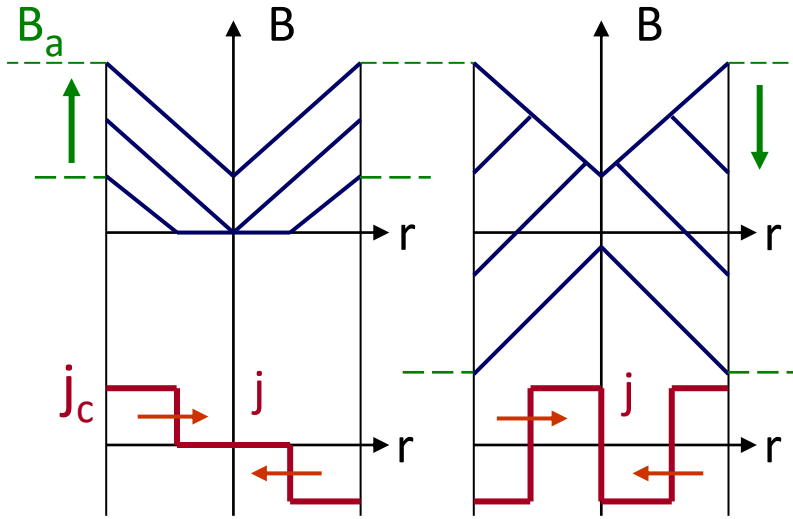
These are the kind of measurements that I intend to do in the experimental part of this course

Methods to measure J_c

2. Magnetization (critical state model, 1964)



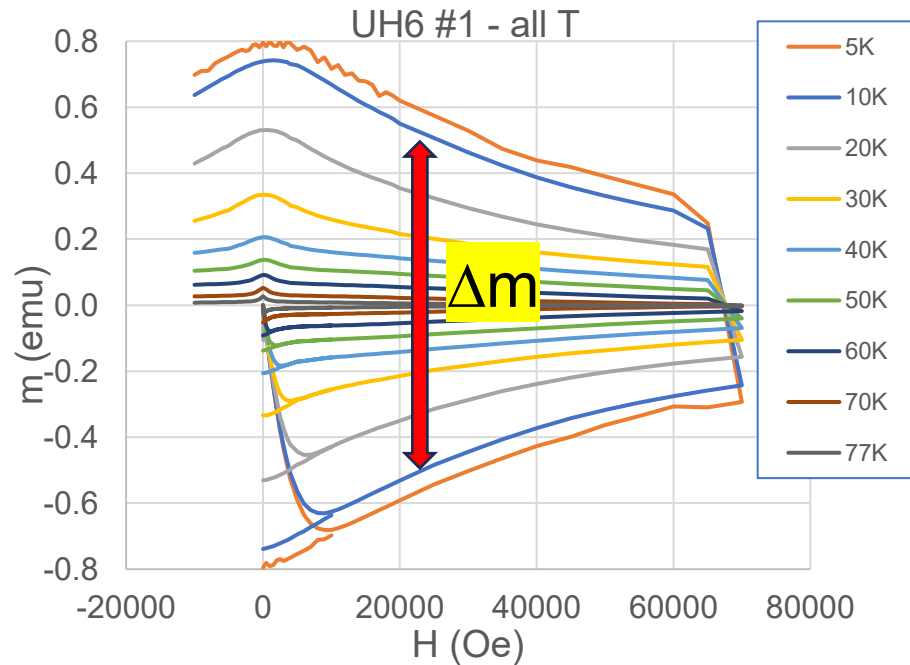
(Simplest case:
long cylinder or slab)



As B_a increases vortices penetrate
But they get pinned near the surface
More vortices push them further in
The vortex system self-organizes in a
“critical state”: $J=J_c$ everywhere, linear
field profile ($J=dB/dx$)

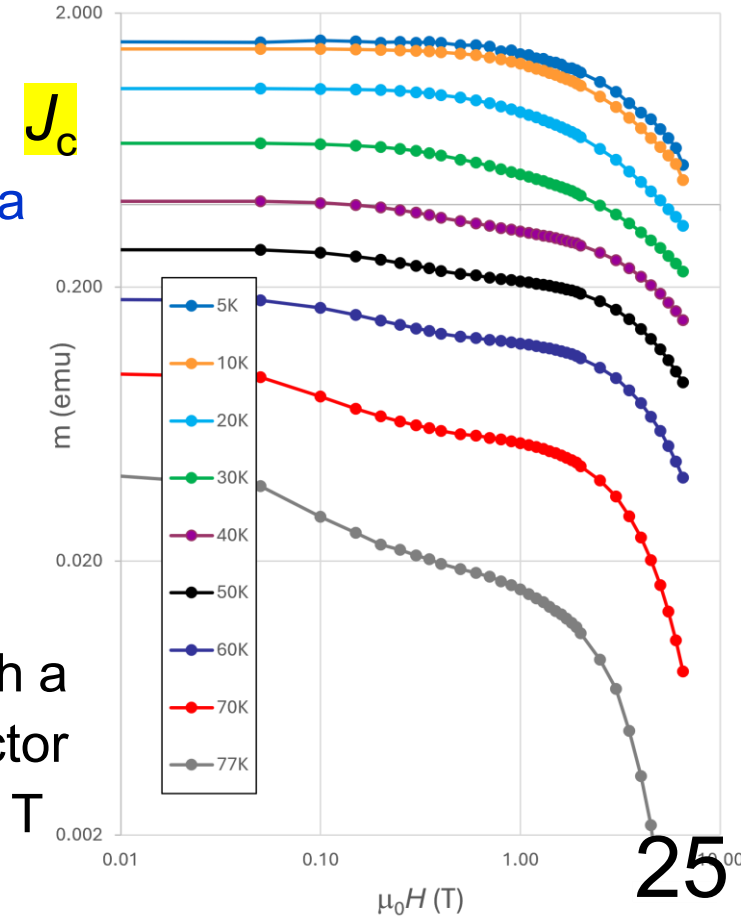


Charles Bean



Our recent data
from MIT on a
CC from Univ.
of Houston

$\Delta m \propto J_c$
Related through a
geometrical factor
independent of T
and B



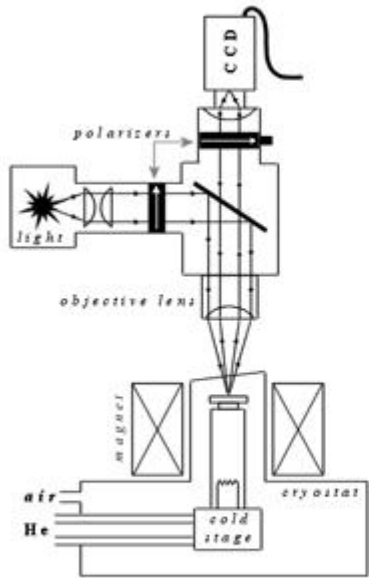
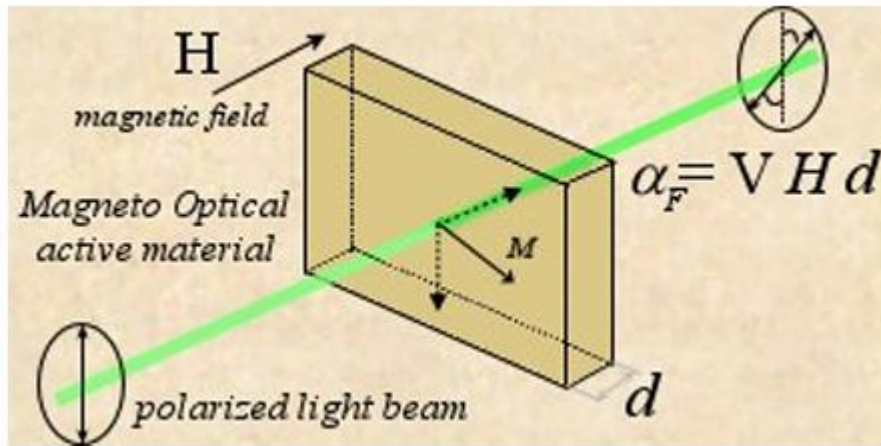
Alexis
Devitre

Zoe
Fisher



Methods to measure J_c

3. Magneto-optics (Faraday effect)

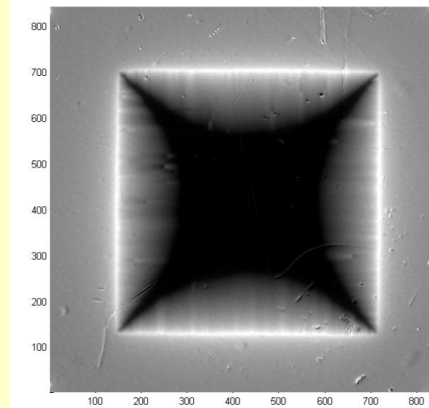


Magneto-optical Lab, Politecnico di Torino

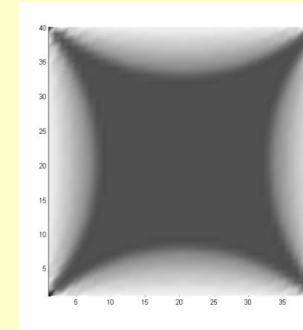


Superconducting YBCO film
(size 1 mm, thickness 300 nm, $T=4K$, $H=3000e$)

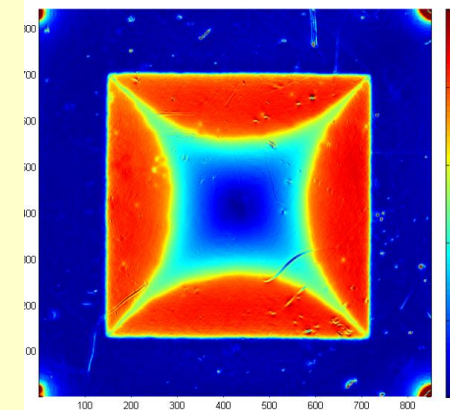
Measured B
distribution



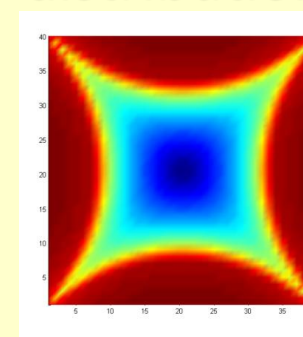
Calculated B
distribution



Measured J
distribution



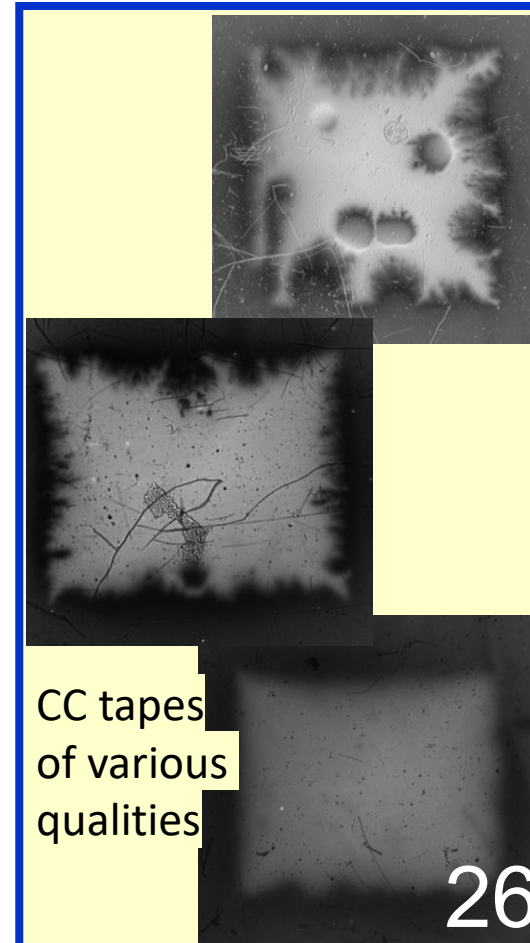
Calculated J
distribution



Francesco
Laviano



Daniele
Torsello



Plan of the class

1. Vortex physics

- Review of the main concepts and results of the Ginzburg-Landau theory
- Single vortex – lower critical field H_{c1}
- Vortex-vortex interactions – Abrikosov lattice – upper critical field H_{c2}
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- Interaction of vortices and electric currents – vortex motion – flux flow – dissipation

2. Vortex pinning

- Origin and importance of vortex pinning - Critical current density
- Classification of pinning centers
- Vortex elasticity – interactions of many vortices & defects – pinning landscape
- Methods to measure J_c

3. Materials for superconducting wires (LTS & HTS)

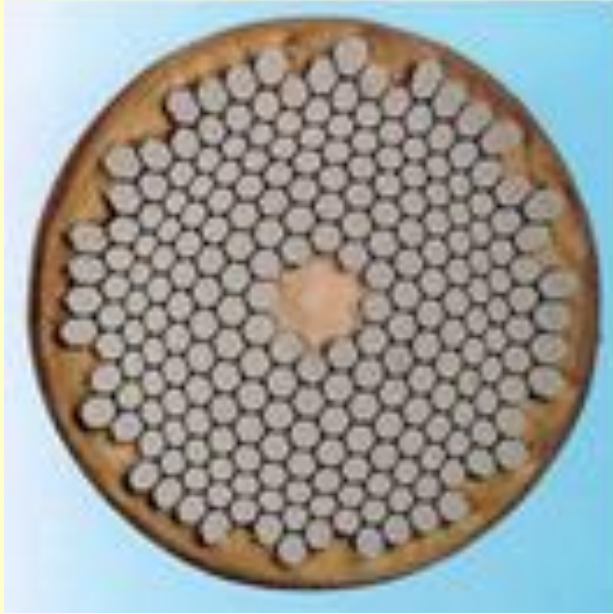
4. Pinning enhancement in ReBCO

- Particle irradiation
- Second phases additions

5. CC for fusion

Materials for superconducting wires: LTS - MgB_2

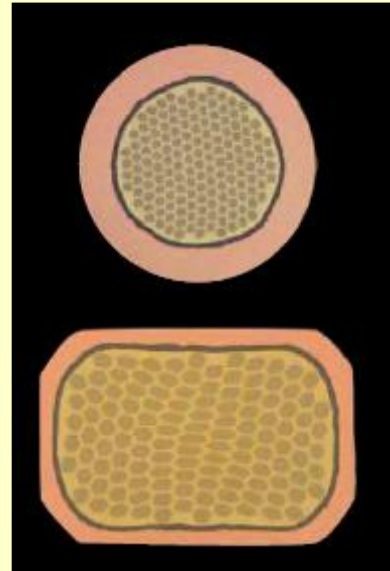
$T_C \approx 9\text{K}$, $H_{C2}(0) \approx 12\text{T}$



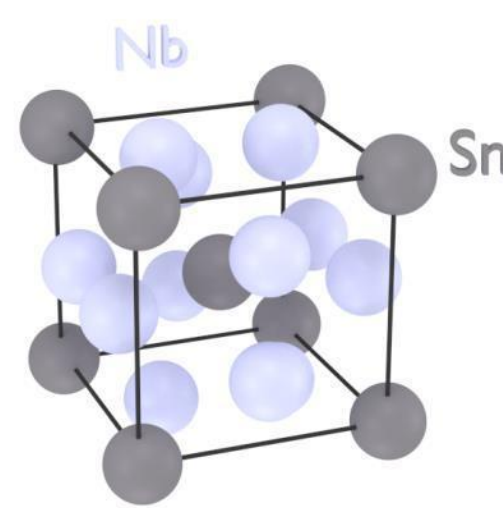
NbTi wire for *LHC magnets*
@CERN (Furukawa Electric):
6300 filaments

Isotropic (cubic)

$T_C \approx 18\text{K}$, $H_{C2}(0) \approx 28\text{T}$



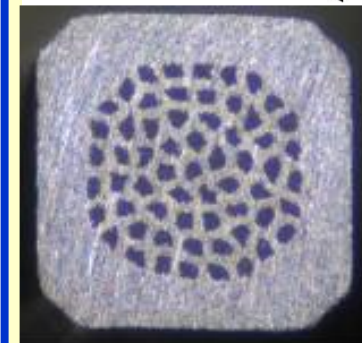
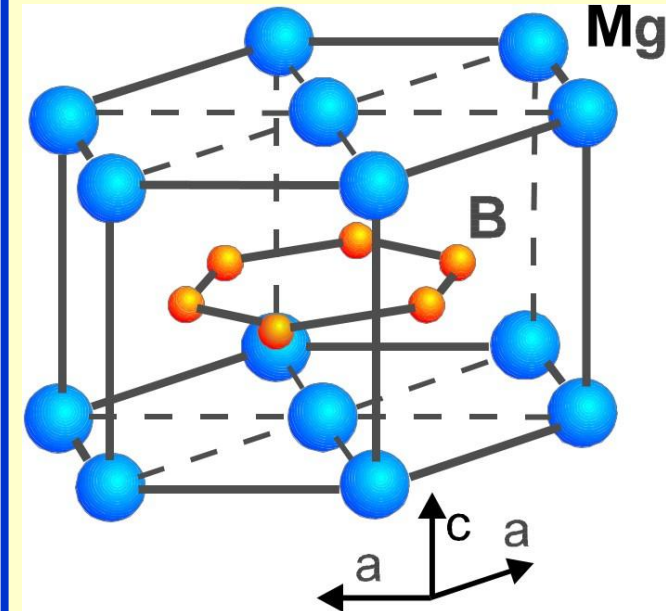
Bruker Nb_3Sn multifilamentary wire,
bronze process



Anisotropic
(hexagonal)

MgB_2

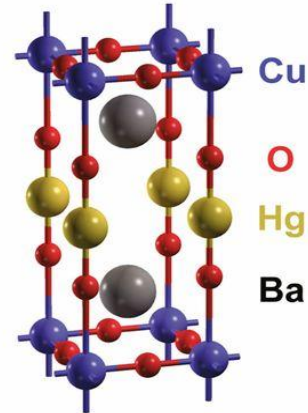
$T_C \approx 39\text{K}$, $H_{C2}(0)$ up to
 $\approx 45\text{T}$ (doped with C)



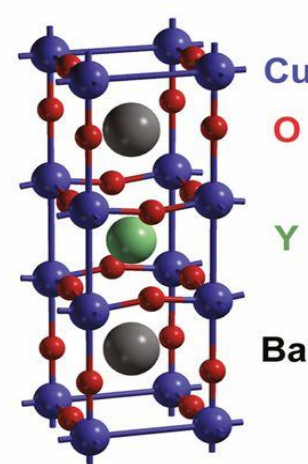
Materials for superconducting wires: HTS Cuprates

- Family of many compounds
- Many with $T_c > 77\text{K}$
- Highly anisotropic (tetragonal or orthorhombic)

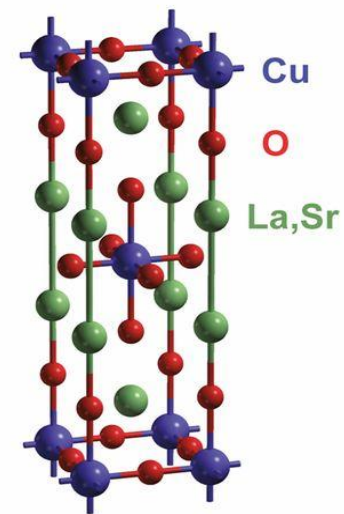
$\text{HgBa}_2\text{CuO}_{4+\delta}$
(Hg1201)



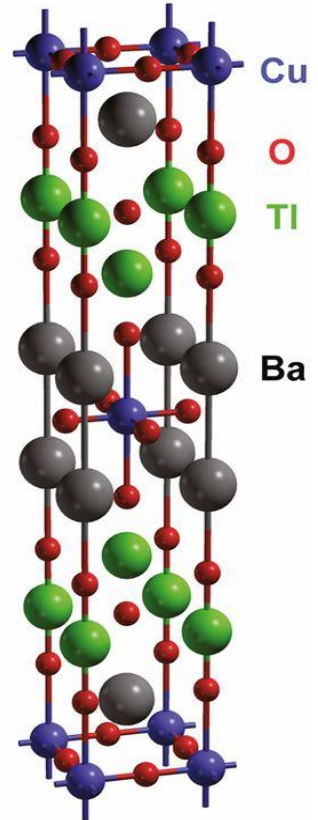
$\text{YBa}_2\text{Cu}_3\text{O}_{6+\delta}$
(YBCO)



$\text{La}_{2-x}\text{Sr}_x\text{CuO}_4$
(LSCO)

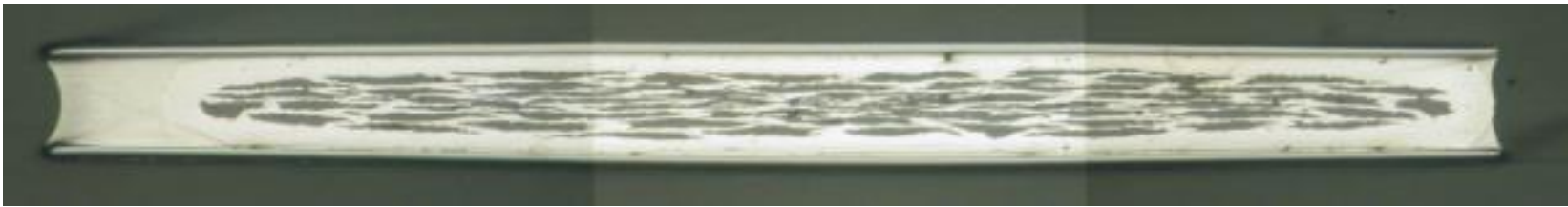


$\text{Ti}_2\text{Ba}_2\text{CuO}_{6+\delta}$
(Ti2201)



Example of Bi-2223 multifilamentary wire

* manufactured by Sumitomo Electric Industries, Ltd.

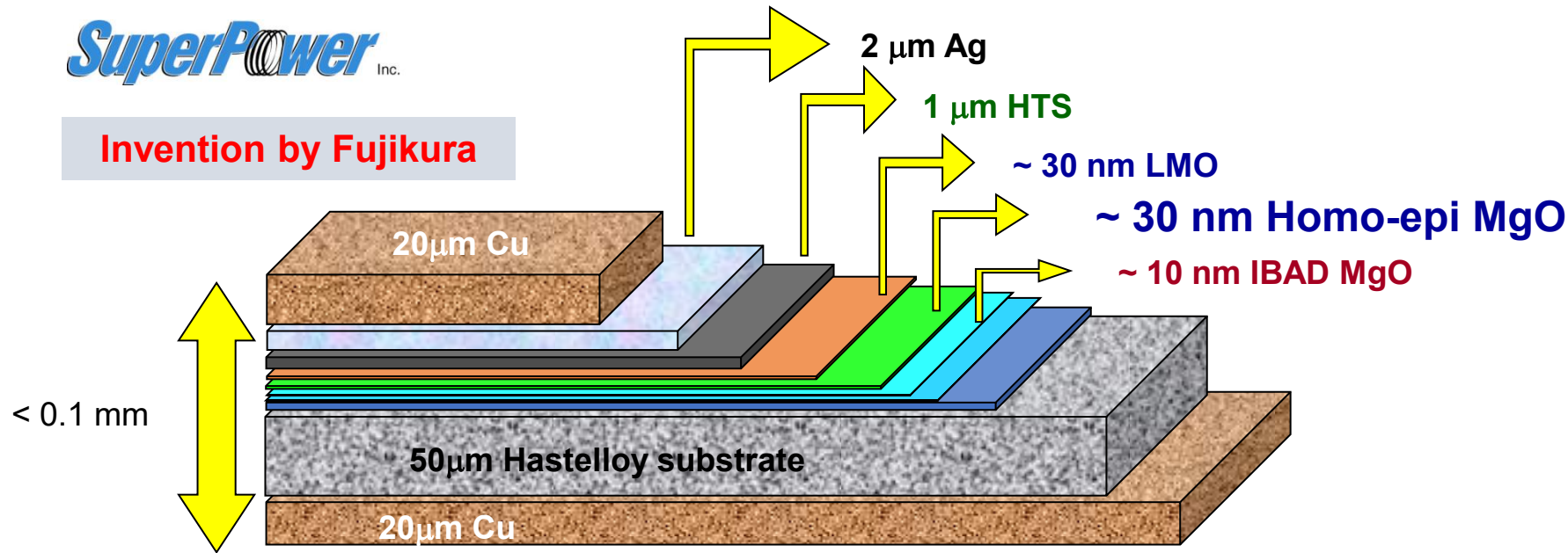


Superconducting (SC) tape	Bi-2223/Ag*
SC tape dimensions	2.5mm ^{wide} x 0.18mm ^{thick}
Number of filaments	55

We will focus on ReBCO coated conductors

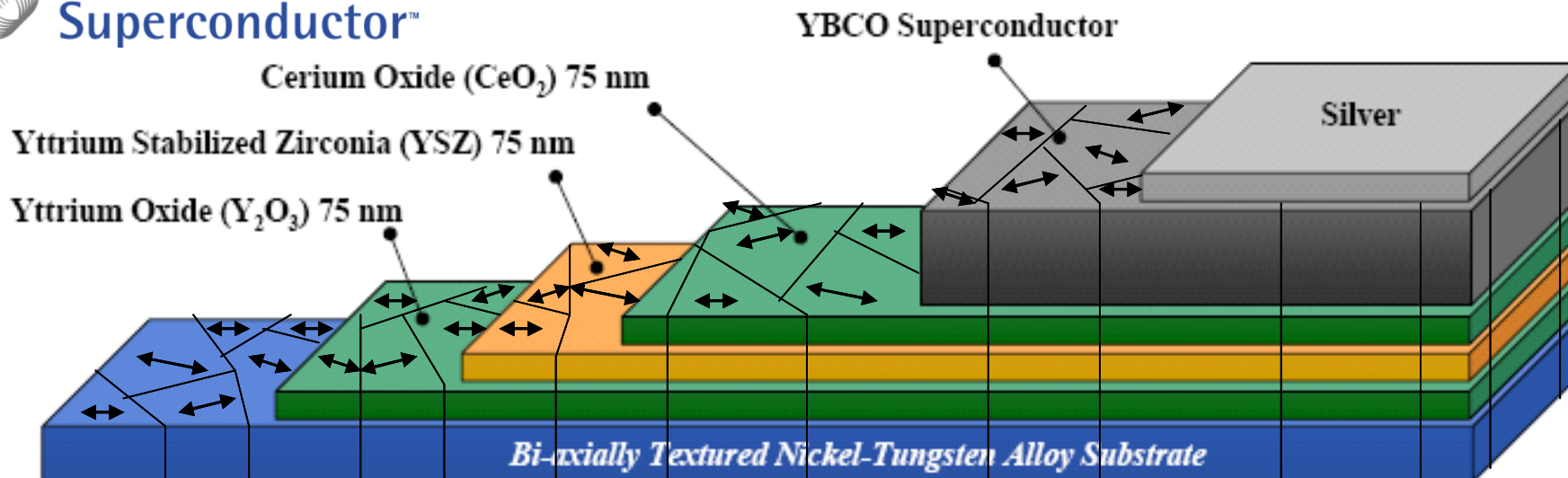
SuperPower Inc.

Invention by Fujikura



Ion-beam-assisted deposition (IBAD) of the textured template

American Superconductor™



Invention by ORNL

Rolling-Assisted, Biaxially Textured Substrate (RABiTS)

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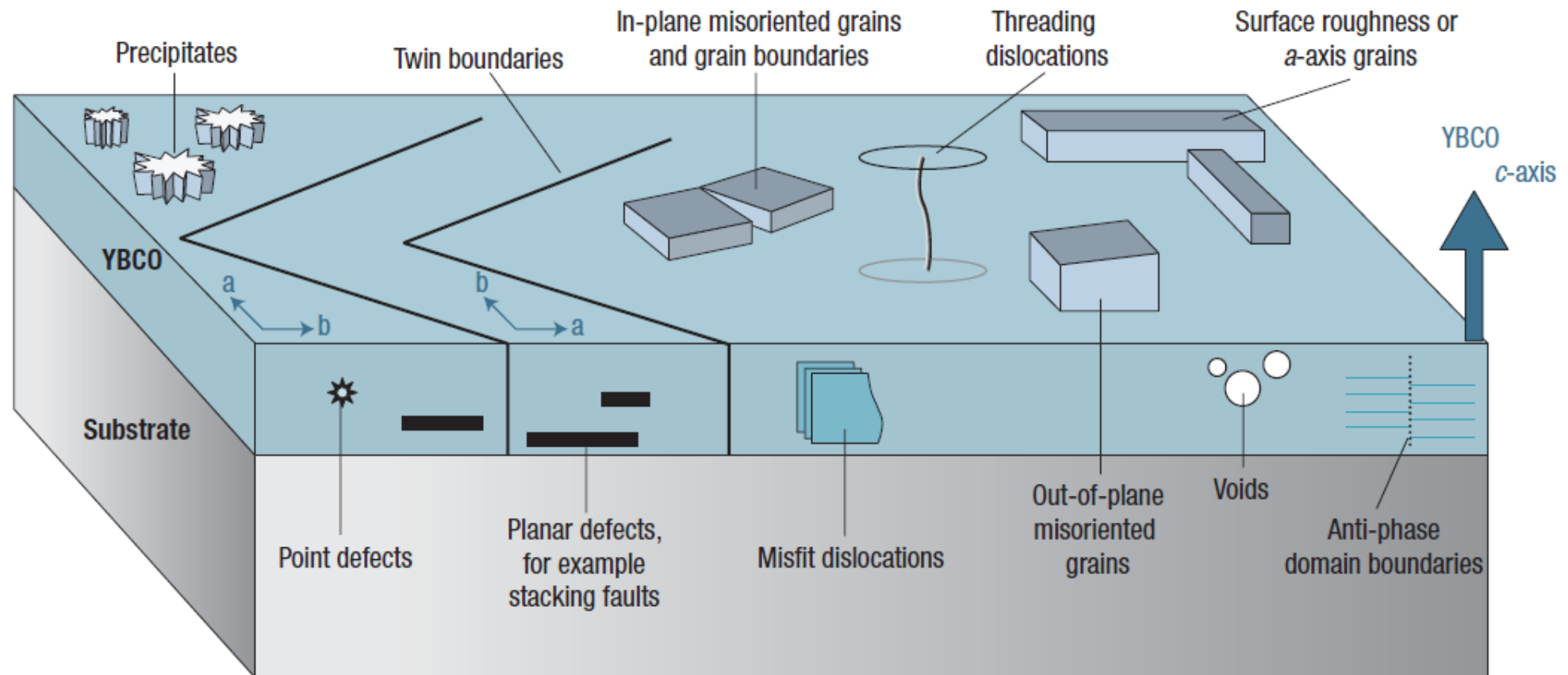
3. Materials for superconducting wires (LTS & HTS)

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- Particle irradiation
- Second phases additions

5. CC for fusion

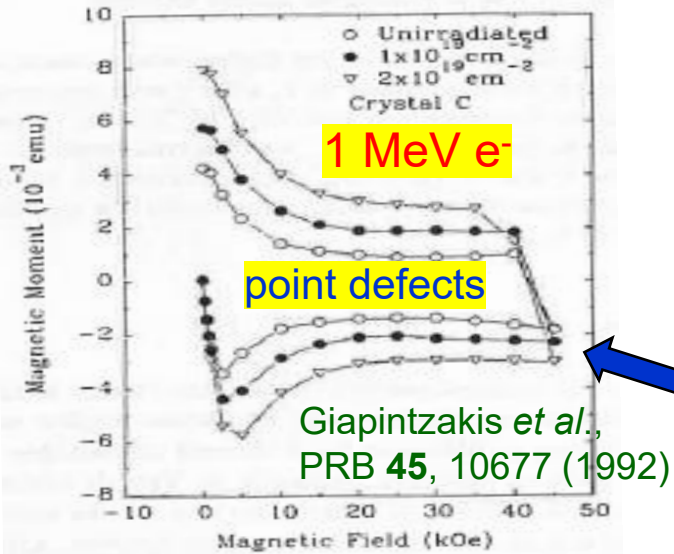
Defects in ReBCO films & CC



Natural defects (formed during fabrication) in ReBCO films and coated conductors
Some of them are specific to the deposition method (laminar or columnar growth)

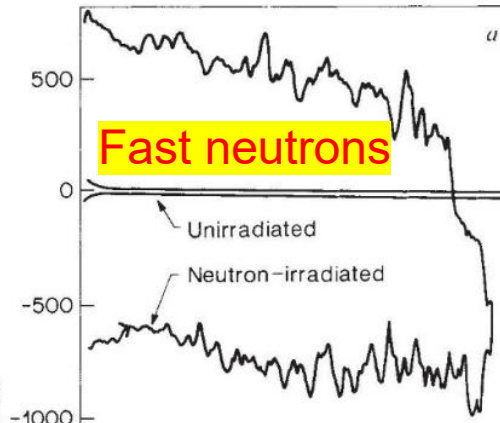
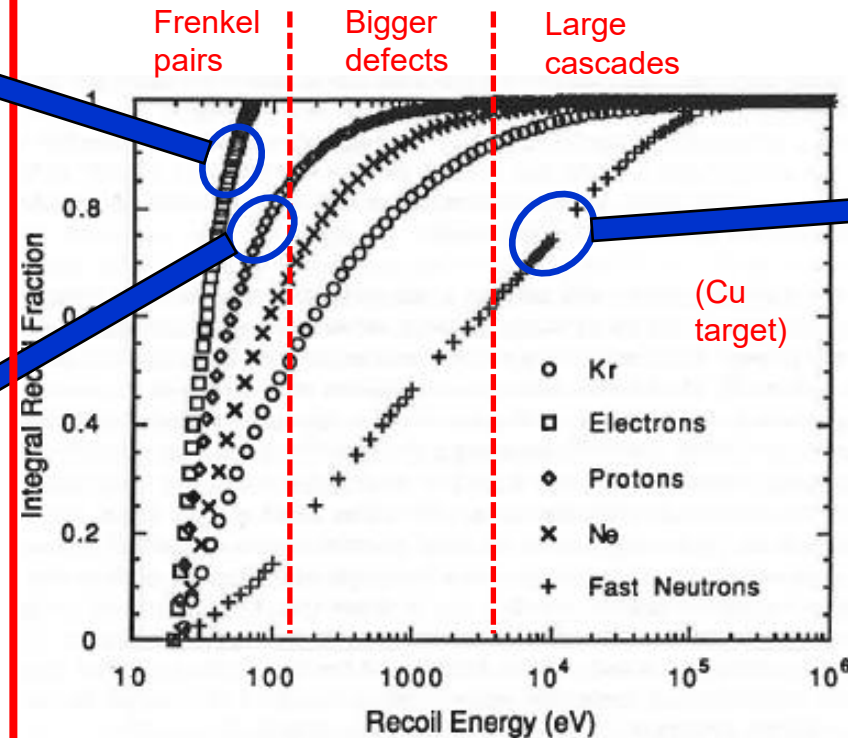
A bit of history:
Particle irradiation of HTS was a popular activity
in the early 1990s (to increase J_c)

Random disorder created by light ions (energy < few MeV) & neutrons



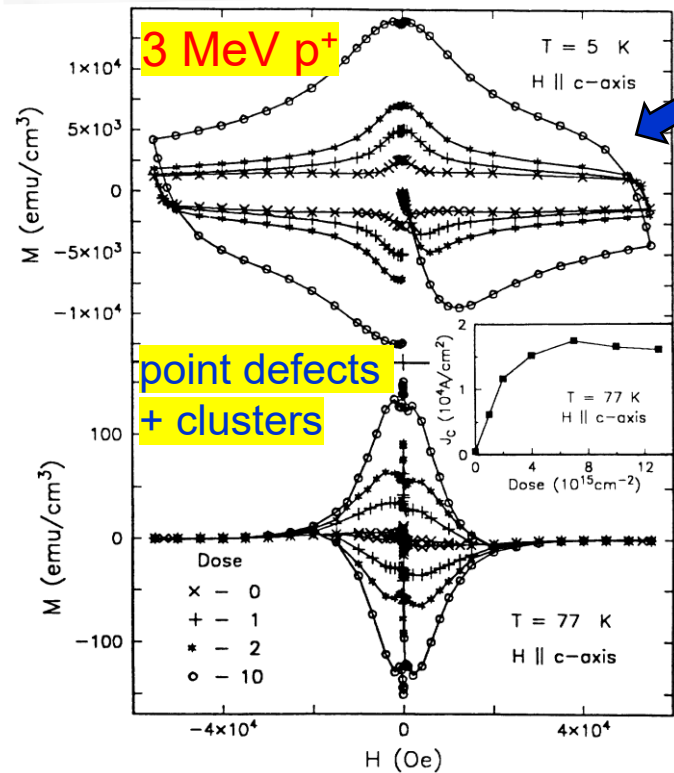
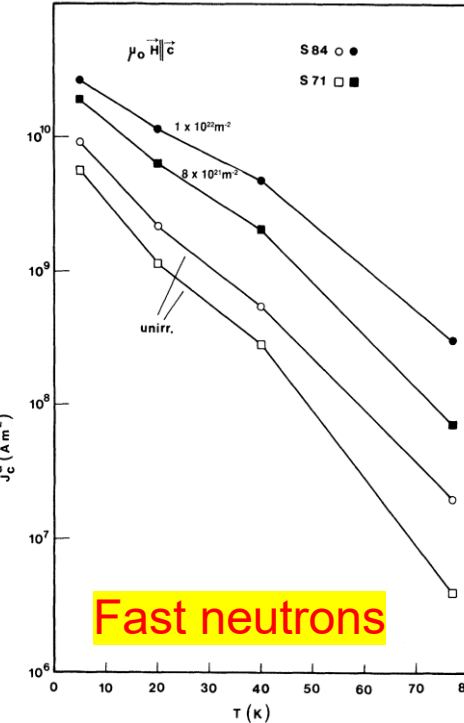
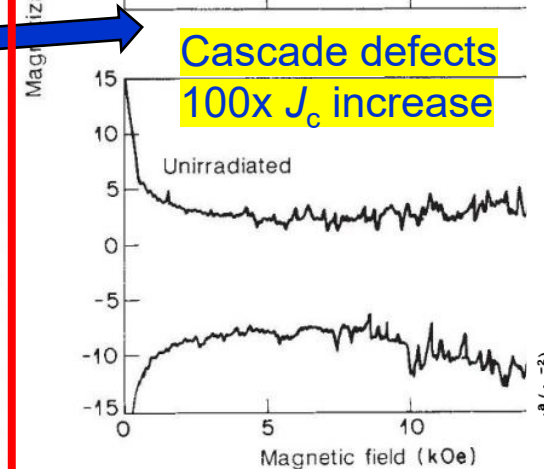
Direct collisions with lattice nuclei
(nuclear or non-ionizing energy loss):
Dominant for light ions up to few MeV

Localized uncorrelated disorder



R. van Dover *et al.*,
Nature **342**, 55 (1989)

F.M. Sauerzopf *et al.*,
PRB **43**, 3091 (1991)



L. Civale *et al.*, PRL **65**,
1164 (1990)

Orders of magnitude increases in J_c
in clean YBCO single crystals

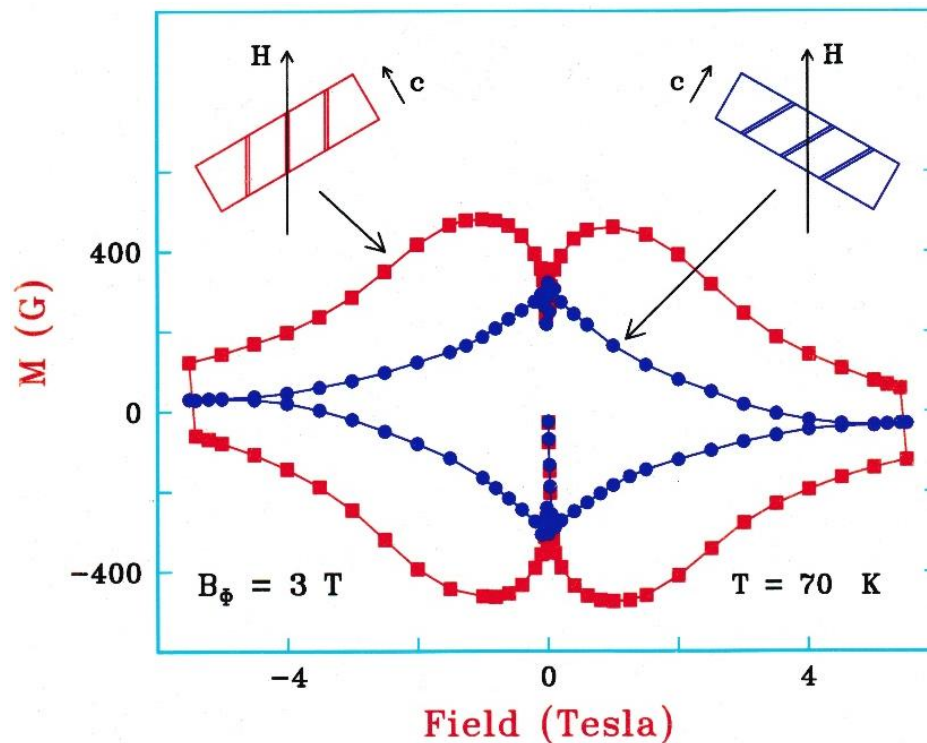
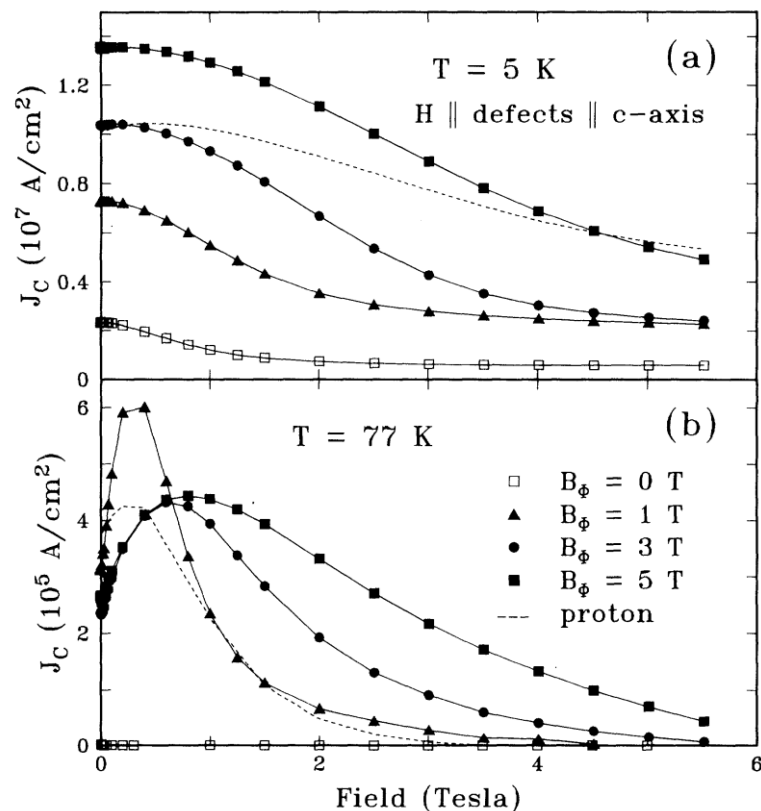
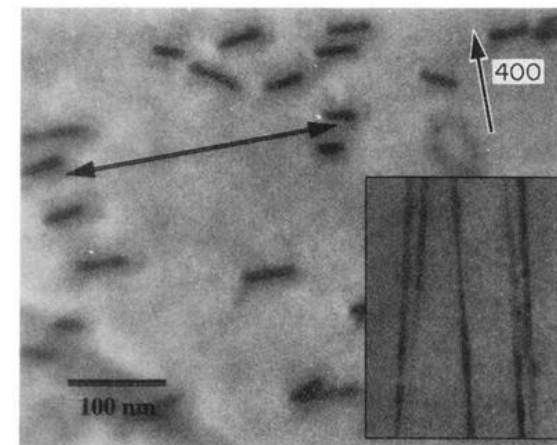
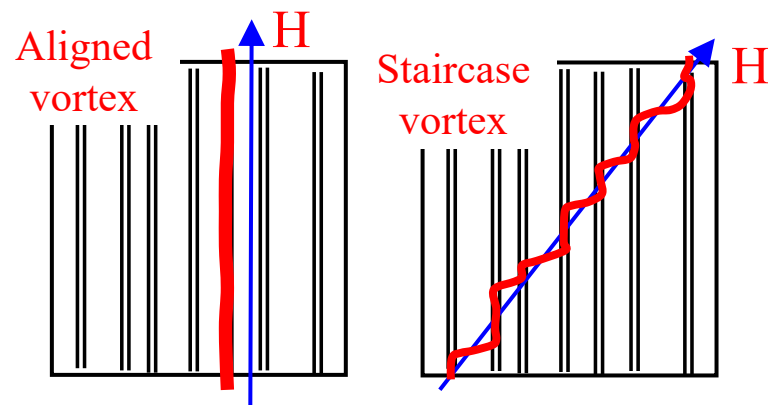
High energy heavy ion irradiation: latent tracks by electronic excitations

Columnar defects (Correlated disorder)

⇒ Directional pinning

⇒ **Most effective pinning for $H \parallel$ defects**

⇒ Much better than random at high T



CD misalignment (splay),
tuned by heavy ion energy,
further improves J_c

L. Civale *et al.*, PRB
50, 4102 (1994)

L. Civale *et al.*, PRL 67, 648 (1991)

J_c in ReBCO films and CC can be increased
by chemical introduction of defects (second phases)

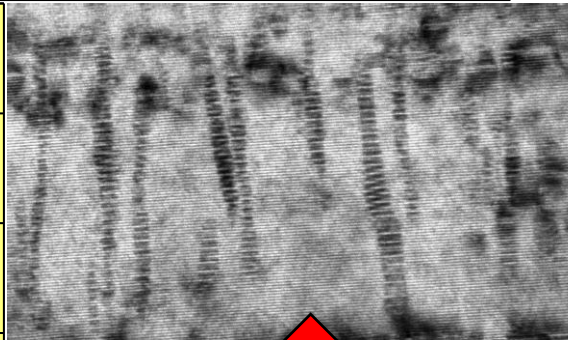
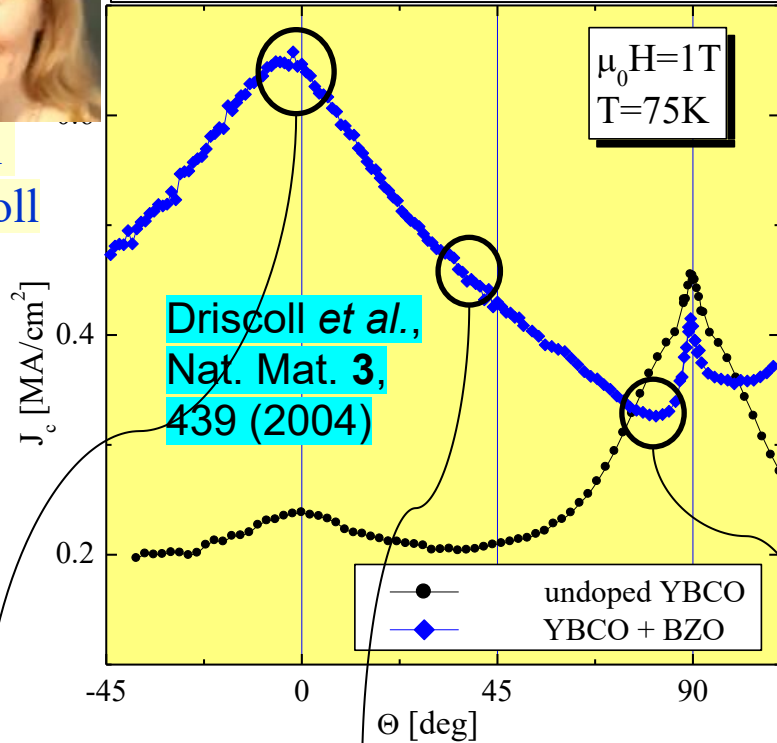
Strong J_c enhancement in PLD ReBCO films & CC by BZO nanorods



Judith Driscoll

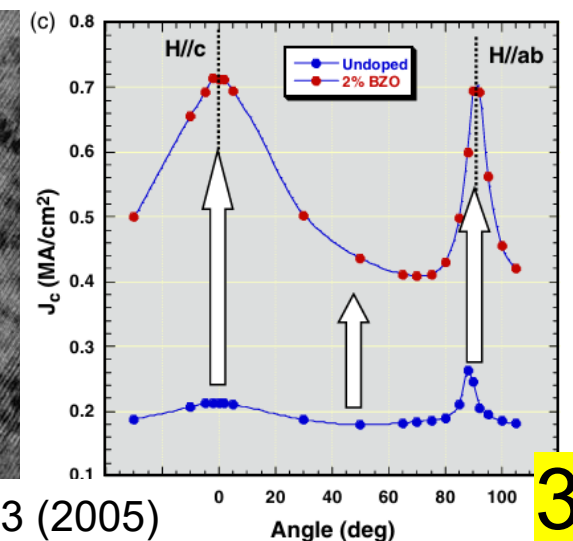
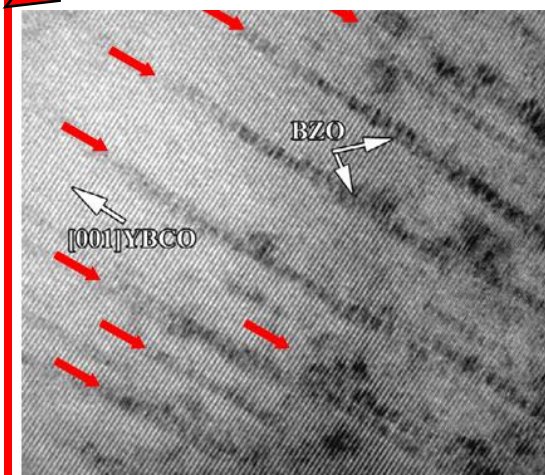
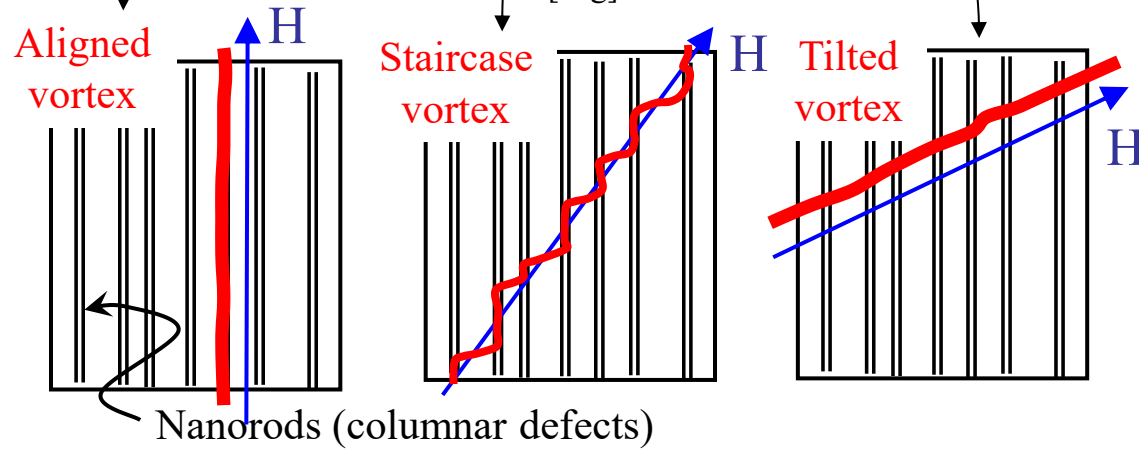
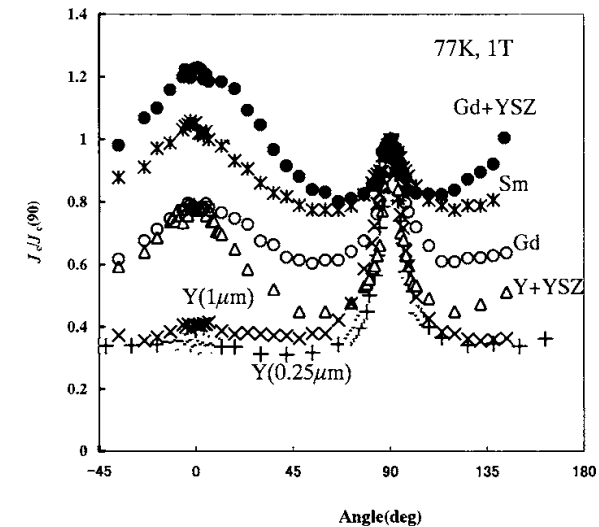
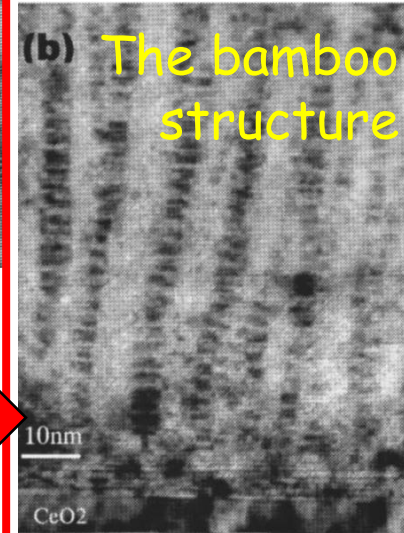
We produced record high J_c in PLD YBCO films by adding BaZrO_3 self-assembled nanorods

This work triggered lots of studies with the same goal of enhancing J_c in ReBCO CC



BaZrO₃ nanorods

Yamada et al. Appl. Phys. Lett. 87, 132502 (2005)



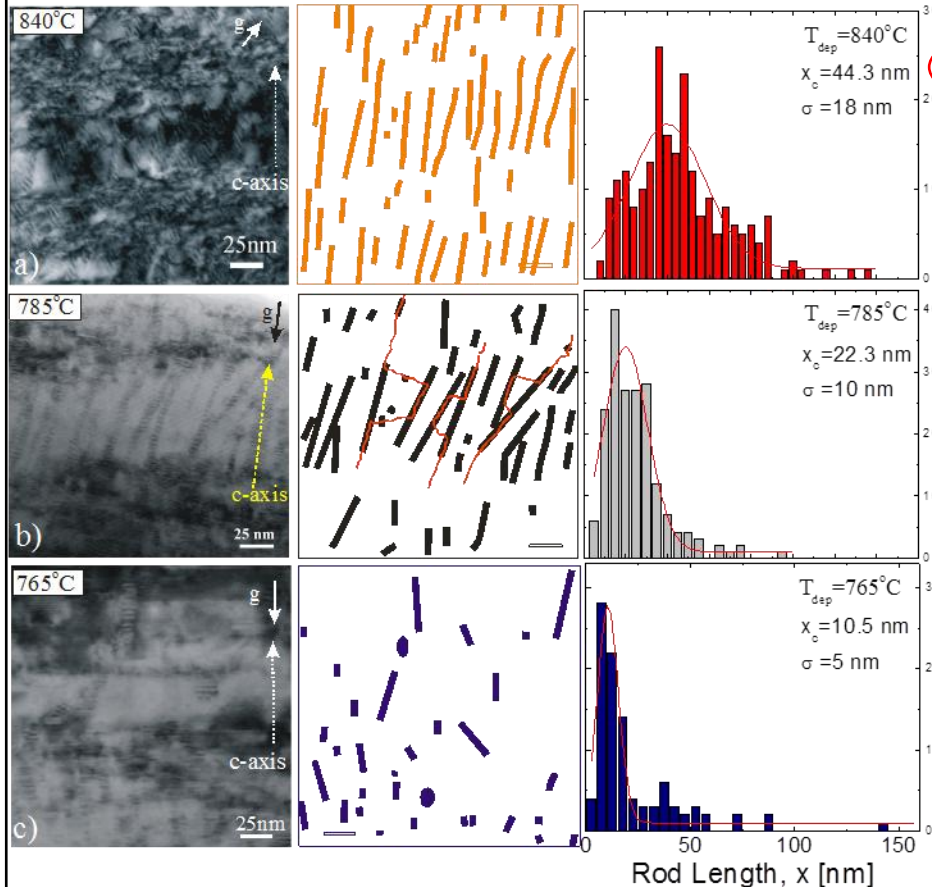
Goyal et al., SuST. 18, 1533 (2005)

The advantage of mixed pinning landscapes

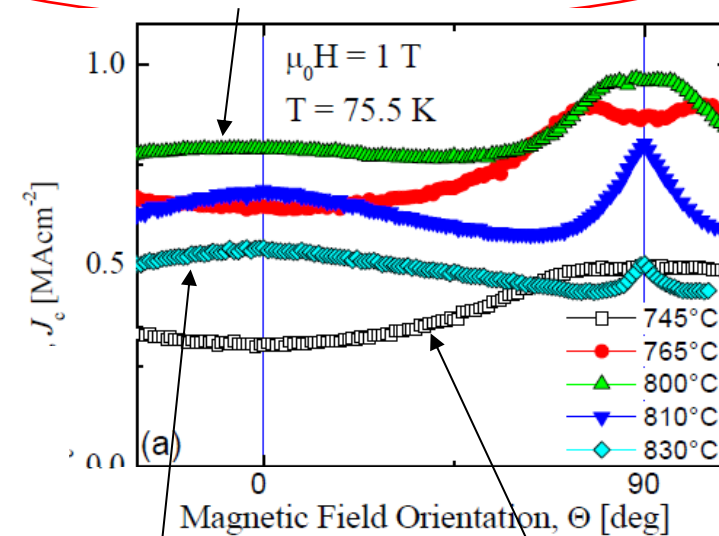
We learned to nanoengineer the pinning landscape by tuning the growth conditions in PLD YBCO+BZO:

Low growth temperature or High rate = random nanoparticles
High growth temperature or Low rate = self-assembled nanorods

Boris Maiorov



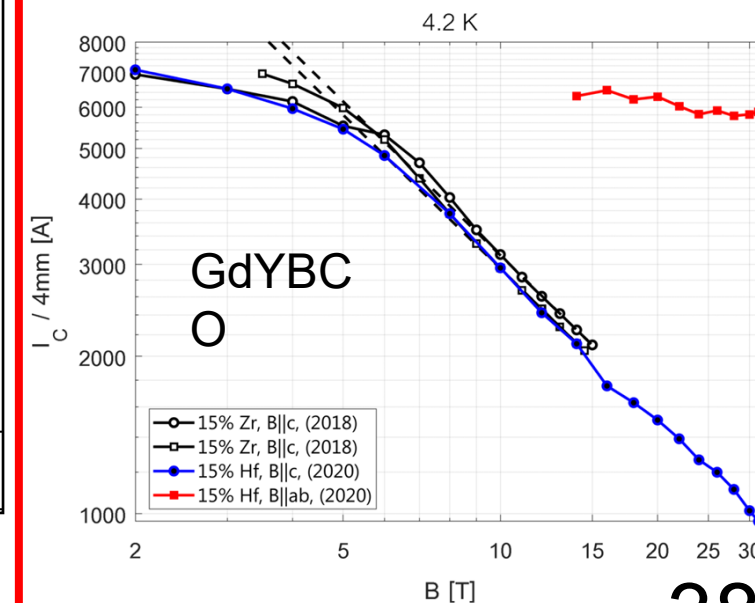
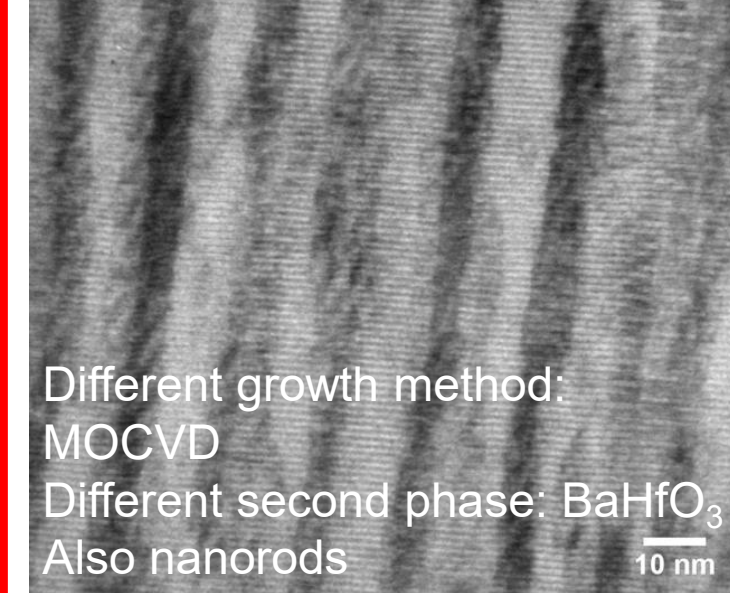
Mixed pinning landscape (splayed nanorods + nanoparticles): best J_c



Columnar defects: large c-axis peak
Random nanoparticles: no c-axis peak

B. Maiorov et al., Nat. Mat. 8, 398 (2009)

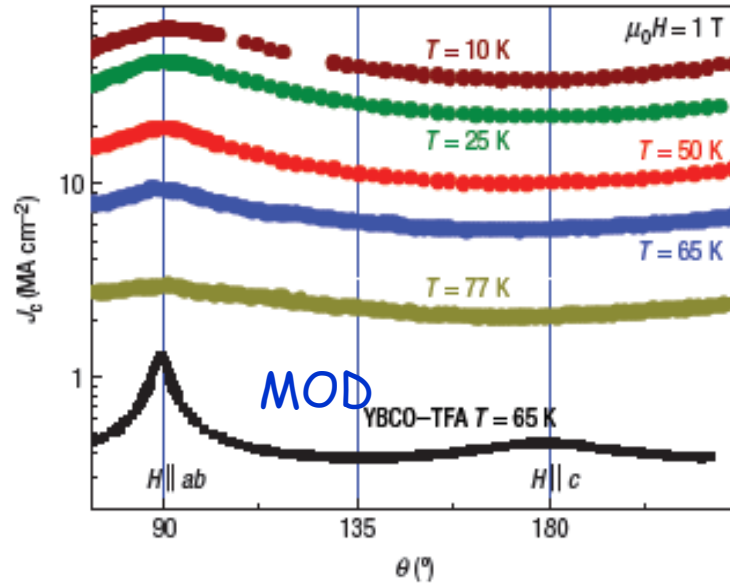
G. Majkic et al. SuST 33 07LT03 (2020)



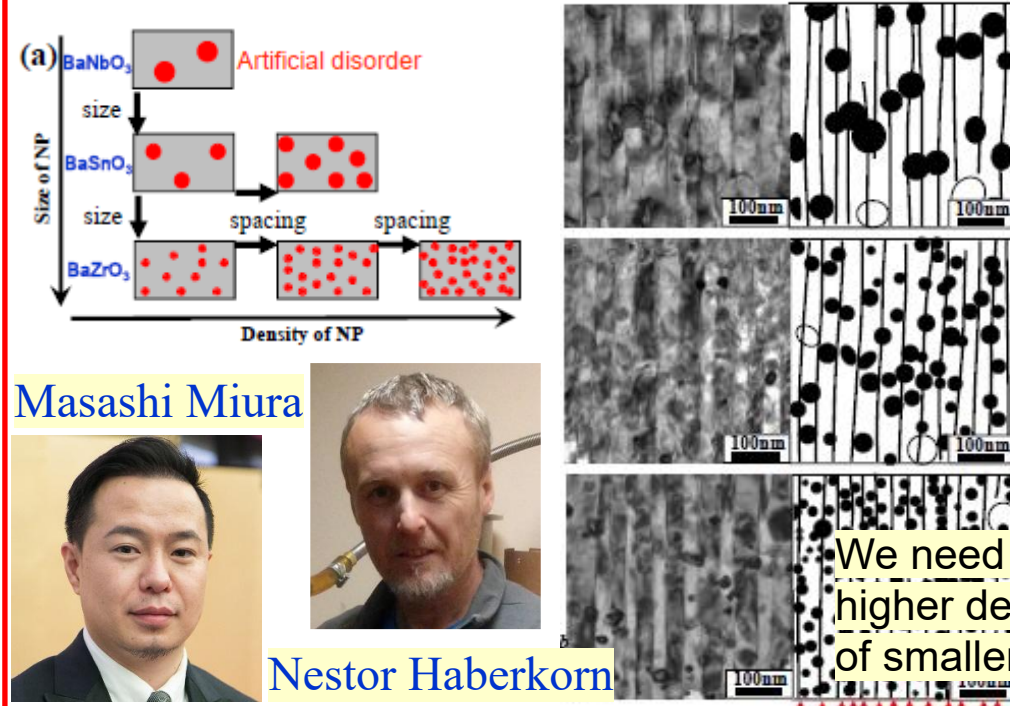
Processing → properties correlations: The columnar growth of Pulsed Laser Deposition (PLD) films promotes the self-assembly of BZO nanorods

Second phases in CC grown by MOD: strong pinning by random nanoparticles

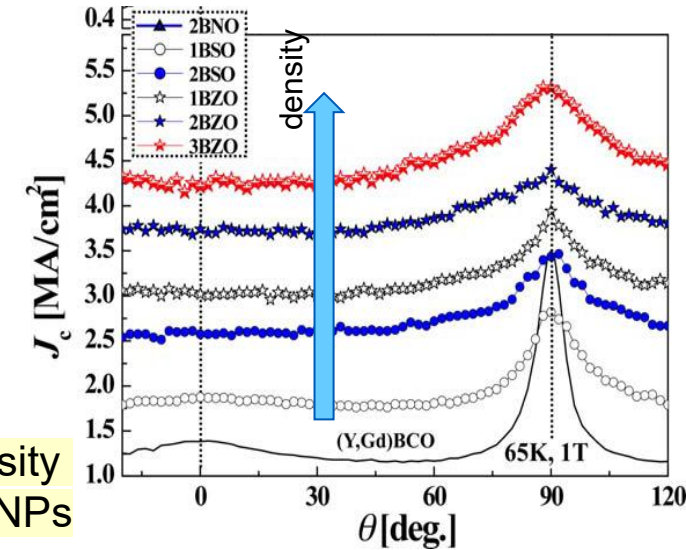
J. Gutierrez *et al.*, Nat Mat. **6**, 367 (2007)



- Strong pinning by nanoparticles
- no c-axis correlated disorder

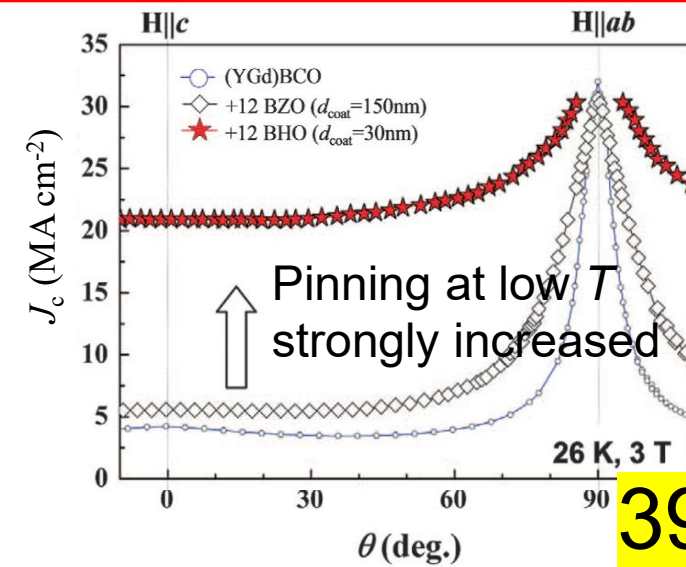
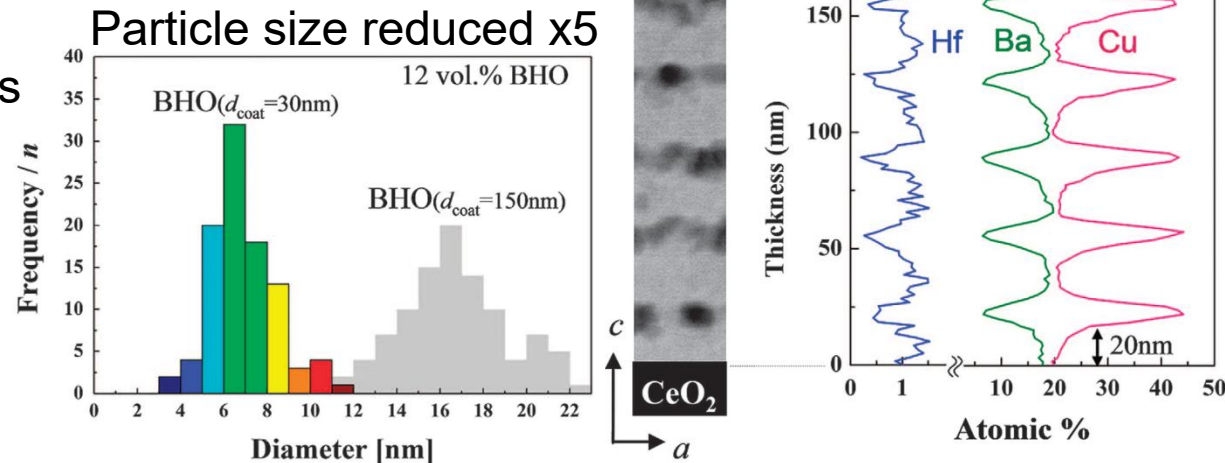


M. Miura *et al.*, PRB **83**, 184519 (2011);
SuST **26**, 035008 (2013) - N. Haberkorn
et al., Phys. Rev. B, **85**, 174504 (2012).



Multilayer deposition creates Ba-poor regions that stop the BaHfO_3 NPs growth

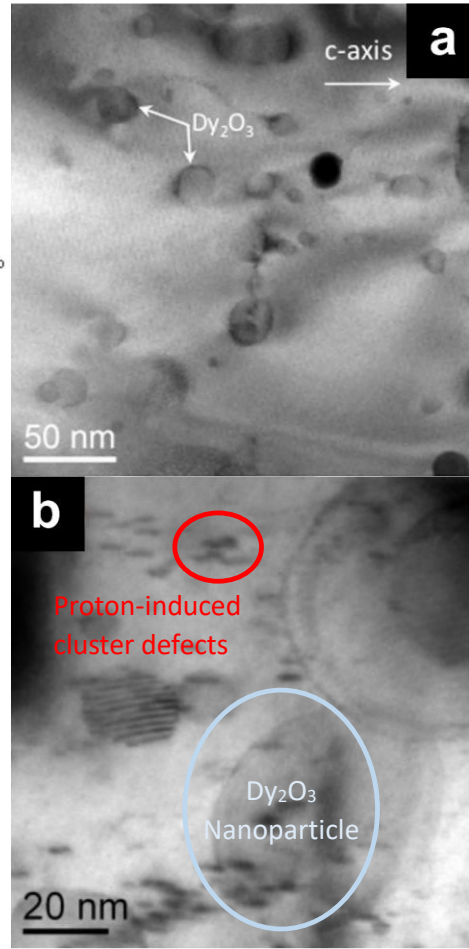
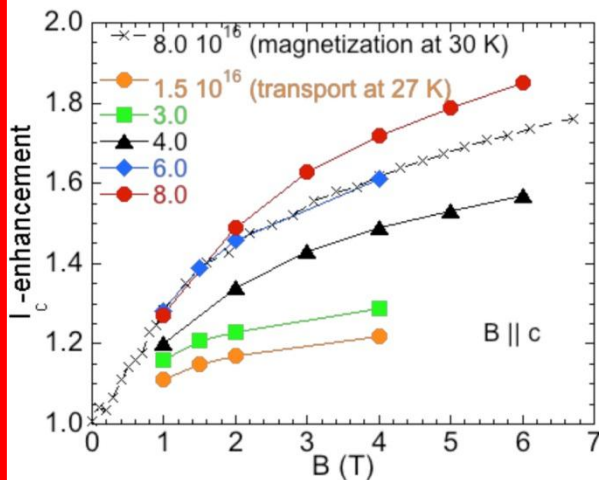
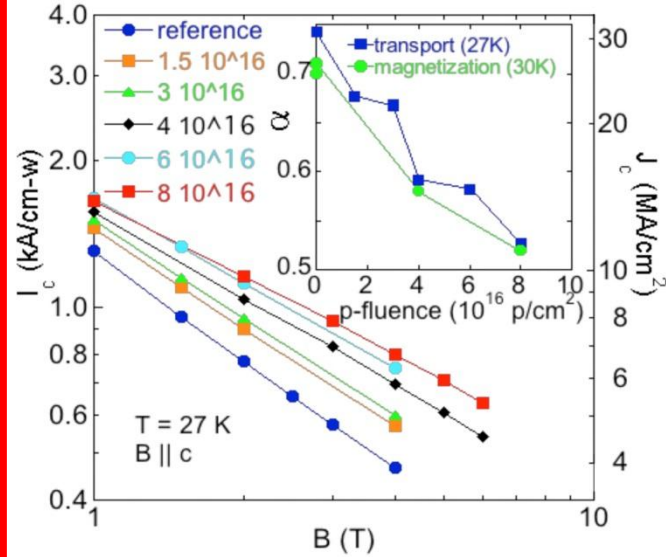
Miura, Maiorov, *et al.*, NPG Asia Materials **9**, e447 (2017)



Pinning in commercial CC can still be enhanced by irradiation

(Much more modestly than in single crystals)

- MOD CC from AMSC
- Irradiation: 4 MeV protons



Y. Jia *et al.*, Appl. Phys. Lett. **103**, 122601 (2013).

Columnar defects added by 250 MeV Au irradiations

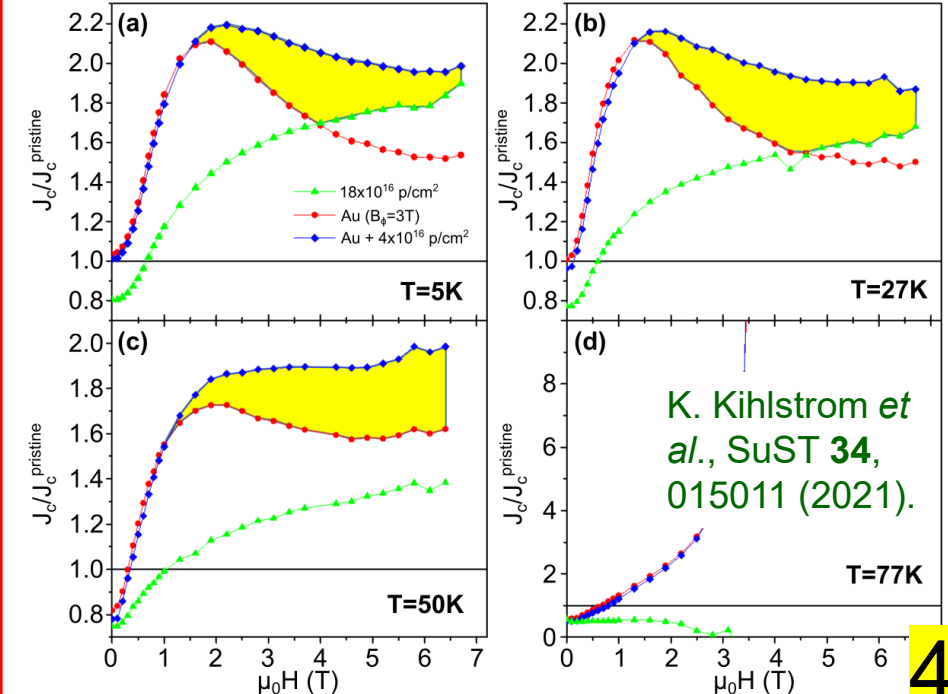
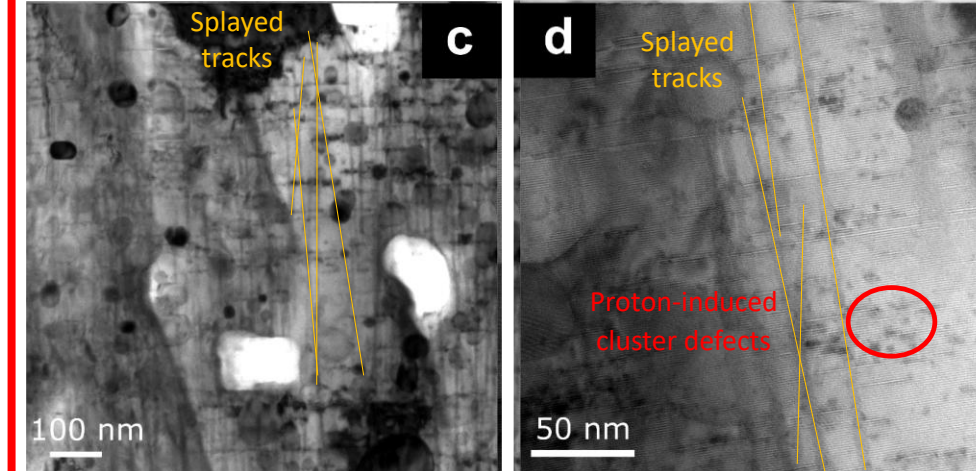


Maxime Leroux



Serena Eley

Combined irradiations: confirmation of the benefits of mixed pinning landscapes



K. Kihlstrom *et al.*, SuST **34**, 015011 (2021).

Status after 2 decades of research
by many groups worldwide:

- ✓ ReBCO is the most studied superconductor, by far
- ✓ Deep understanding of the pinning mechanisms and the processing/properties correlations
- ✓ Many ways to nanoengineer the pinning landscape
- ✓ ReBCO films and CC have the highest J_c of any known SC, by a factor of 5!
- ✓ Mature technology

Plan of the class

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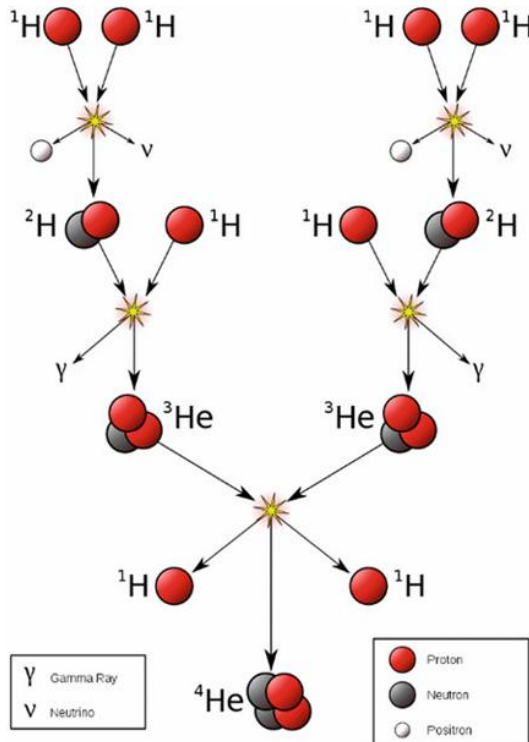
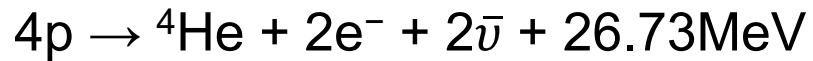
- Particle irradiation
- Second phases additions

5. CC for fusion

Nuclear fusion energy

1920: Arthur Eddington suggests that fusion is the power source of the stars

1939: Hans Bethe identifies the “proton-proton” chain as the main (initial) fusion process in the stars:

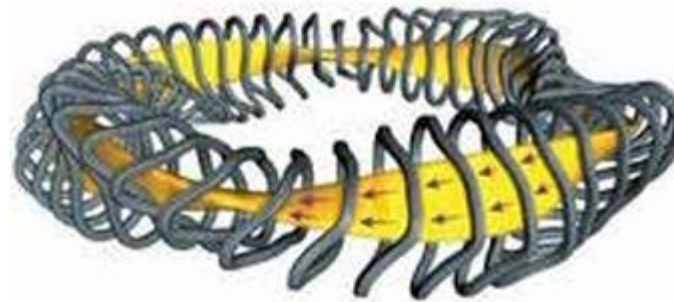


To produce controlled fusion on Earth, we need to create and confine plasmas at huge temperatures ($\sim 10^8\text{K}$)

Magnetic confinement

Inertial confinement

STELLARATOR



Lewis Spitzer 1951

TOKAMAK



Lev Artsimovich 1965
(Sakharov & Tamm)

Cheap, virtually unlimited

The most accessible fusion reaction is Deuterium-Tritium

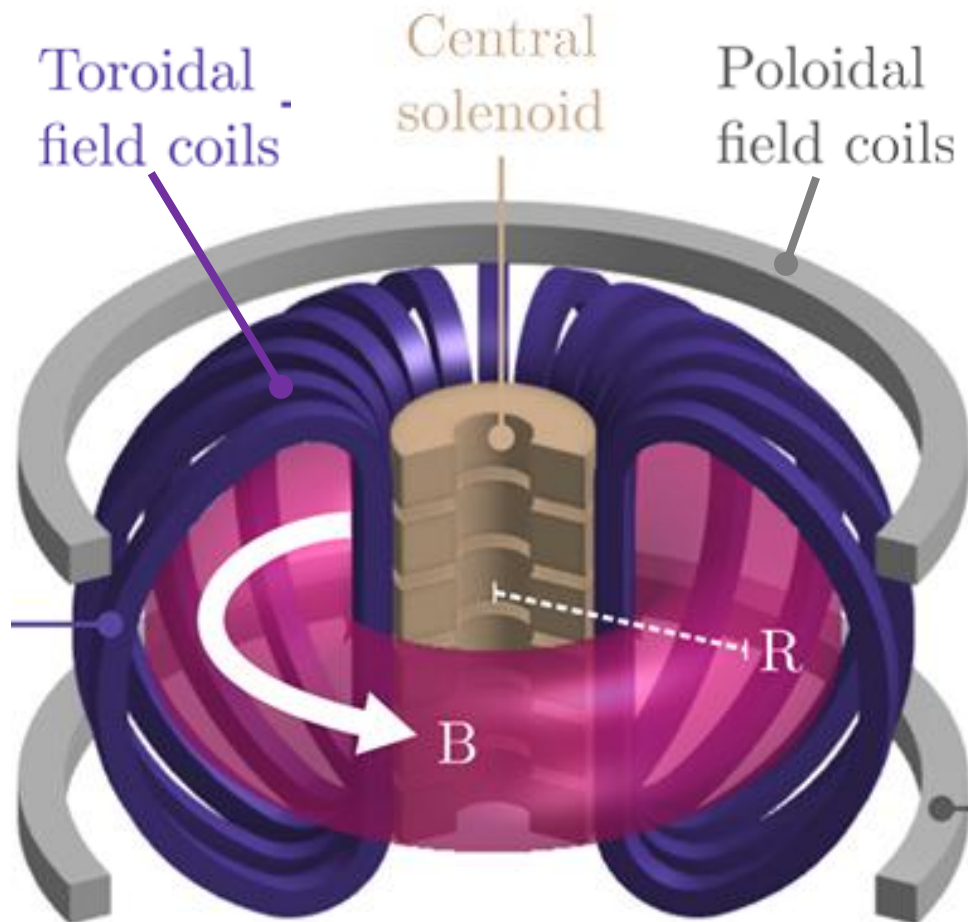


Energy output

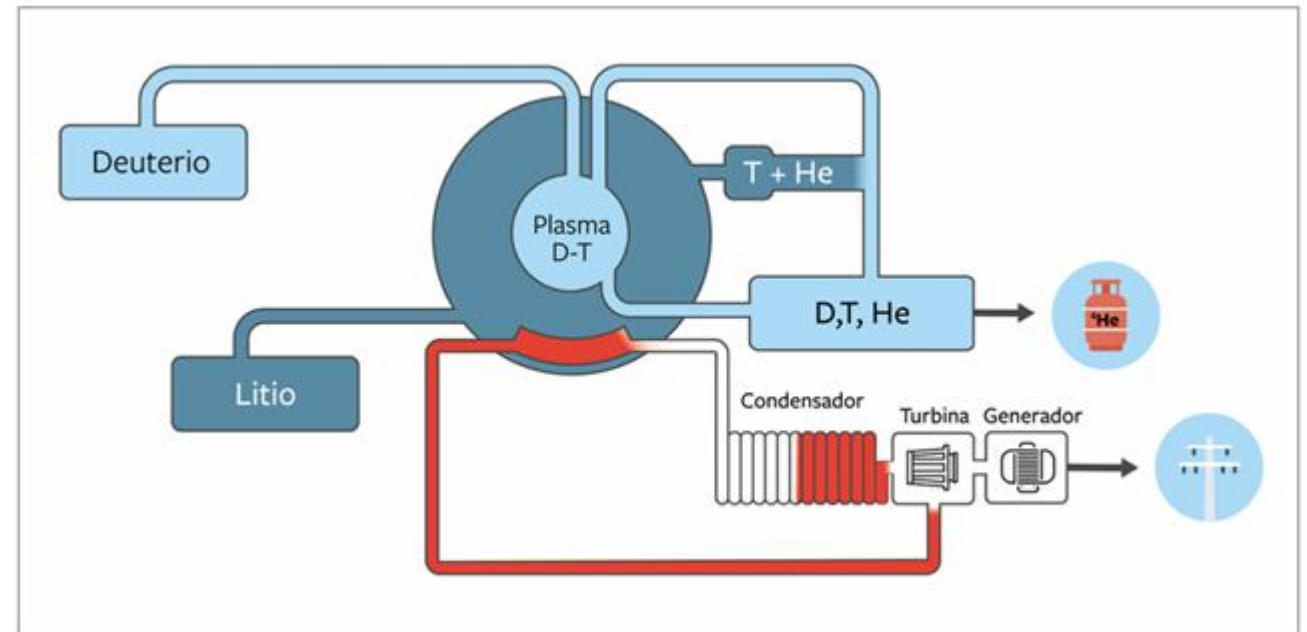
$$T_{\text{ignition}} = 4.8\text{KeV} \sim 5.6 \times 10^7\text{K}, T_{\text{optimum}} \sim 15\text{KeV} \sim 1.7 \times 10^8\text{K}$$

Widespread access to fusion energy would change the world.

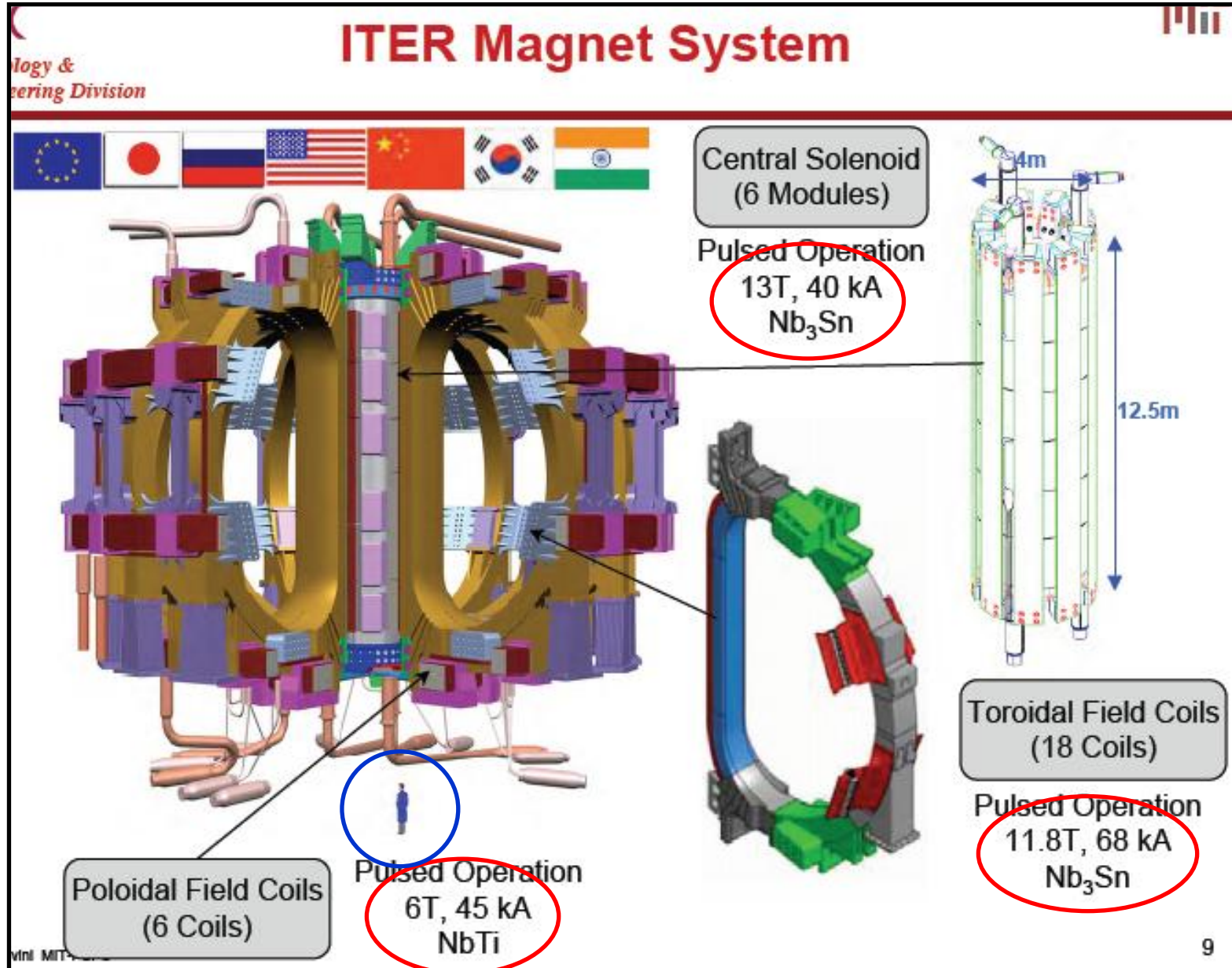
Magnetically-confined fusion reactors **require** SC coils to contain the plasma (e.g. Tokamaks)
Standard option: LTS Nb-alloys (have been used for decades in MRI & other applications).



Clearly disruptive technology.

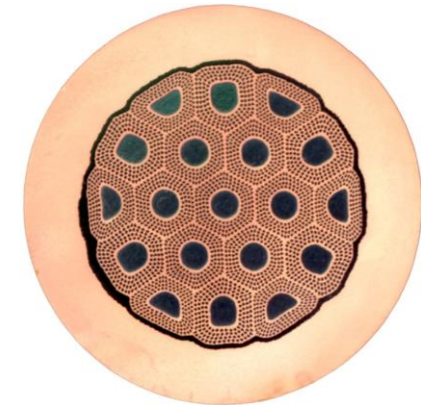


ITER: State-of-the-art of magnetically confined fusion with LTS



The goal of ITER is to achieve fusion power production at power plant scale, breaking new ground in fusion science and demonstrating fusion reactor technology (www.iter.org)

Not a power plant yet!!

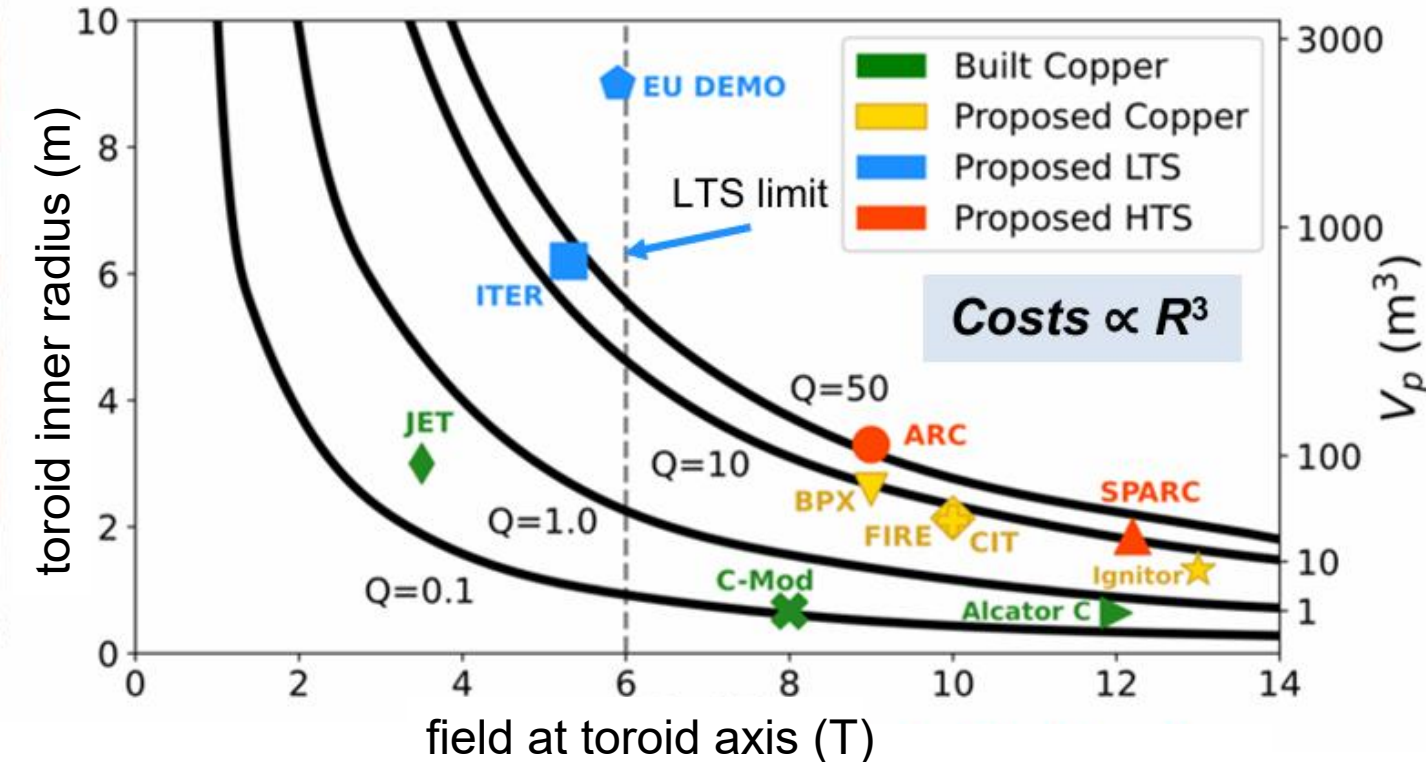


Nb_3Sn strand from Oxford Superconducting Technologies for ITER

Compact fusion reactors magnets: a great opportunity for high HTS

Commercial nuclear fusion power plants have been 20 years in the future.... for several decades now!

But this time, it may be for real!



Huge potential market,
and there is only one option
for the magnets: ReBCO-
based Coated Conductors

- ✓ Critical temperature, T_c – **OK!**
- ✓ Critical current density, J_c – **OK!**
- ✓ Critical magnetic field(s), H_c – **OK!**

Reactor efficiency $\propto B^3$

Big push to build **compact** fusion reactors:

- Operation at $B \sim 20\text{T}$
- Smaller $\Rightarrow$ cheaper ($\sim 100\text{s of M\$}$)
- Driven by private companies, not governments
- Timelines of years instead of decades

Do we have the CC tapes to make the tokamaks?
YES!

- At present, CC with the required properties are routinely made in km-long pieces.
- Supply may be a bottleneck. It is being driven by fusion demand!

Developments at Commonwealth Fusion Systems - MIT

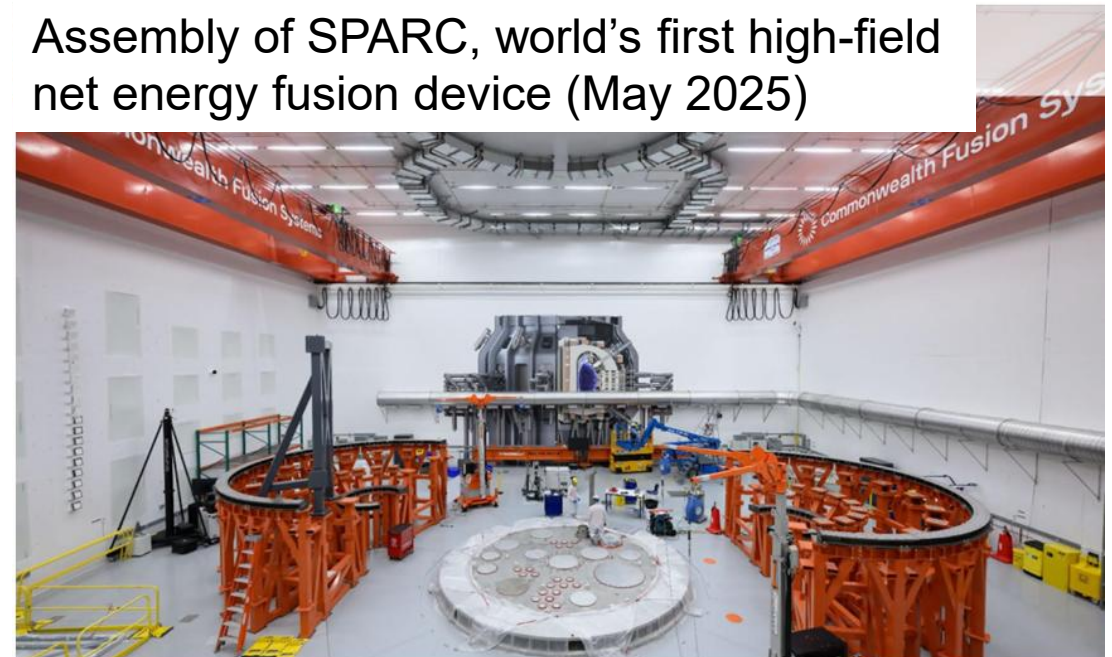


An early CFS coil
(Photo November 2024)



Planned
for 2030s

Assembly of SPARC, world's first high-field
net energy fusion device (May 2025)



Fusion Industry Association (FIA) report:

In 2024, 45 private companies committed to fusion



 **Proxima
Fusion**
<https://www.proximafusion.com/>



<https://www.gauss-fusion.com/>



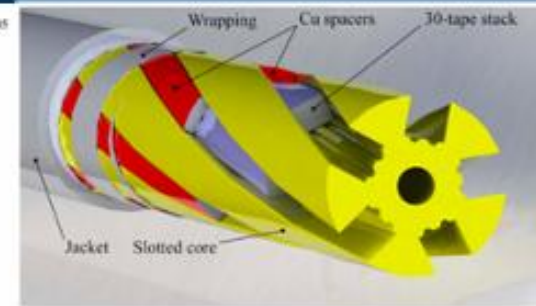
IEEE TRANSACTIONS ON APPLIED SUPERCONDUCTIVITY, VOL. 24, NO. 3, JUNE 2014

Back in 2013/14:

Design of an Industrially Feasible Twisted-Stack HTS Cable-in-Conduit Conductor for Fusion Application

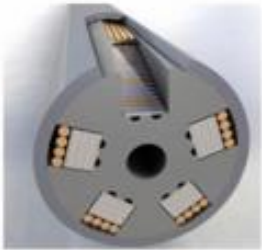
G. Celentano, G. De Marzi, F. Fabbri, L. Muzzi, G. Tomassetti, A. Anemona, S. Chiarelli, M. Seri, A. Bragagni, and A. della Corte, *Senior Member, IEEE*

4601805



Industrial feasibility constantly kept as a design driver

CICC Concept evolution:

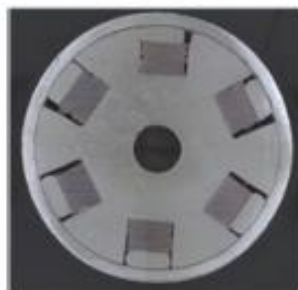


High-strength Al-alloy / steel jacket.
L = 36 mm

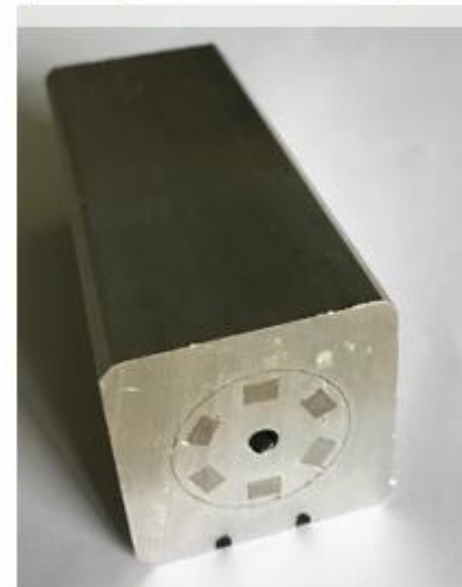
30 – 35 kA current in 18T
(likely for DTT CS)



5-slots
20 kA-class



6-slots
30 kA-class



6-slots core
for improved cooling
(pressure relief channels)



Radiation damage

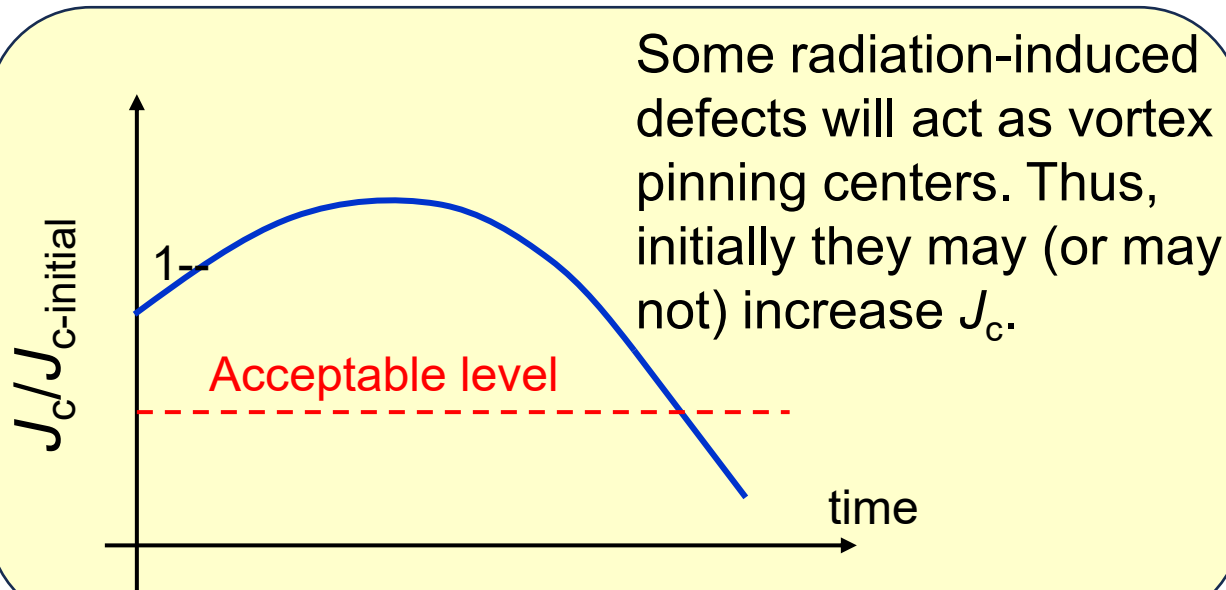
Fusion reactors will generate intense radiation:

- Primary radiation: 14.1 MeV neutrons produced by the Deuterium - Tritium fusion. Present in all reactors.
- Secondary particles: will depend on the reactor design (size, geometry, structural & shielding materials).
- In compact reactors the radiation intensity will be higher.

Radiation will produce material damage (defects).

Inevitably, after some time the damage will deteriorate the properties of the ReBCO CC to the point that the magnets will no longer be operational.

THE BIG QUESTION: how long will the CC magnets last?



TO DO LIST:

- We need to understand radiation damage in CC, particularly the deterioration at high doses.
- We need to understand the interactions of radiation-induced disorder with pre-existing defects. (specific to each CC structure).
- Irradiation will occur “in-operando” conditions (low T , high B). How will that affect the damage?
- What happens when the magnets warm up?

The parasite experiment

- All fusion reactors will produce 14.1 MeV neutrons (Deuterium - Tritium fusion)
- Very few sources of 14.1 MeV neutrons exist
- One of them is the Frascati Neutron Generator (FNG) at ENEA facility in Frascati, Italy
- Problem: the flux is extremely low – irradiation to relevant fluences will require months or years
- FNG is in high demand, exclusive use is impossible

Our solution:

- Mount a sample holder permanently
- Almost all neutrons go through, no interference with other studies
- Leave it there for ~ 3 years
- Check T_c , J_c , etc. every ~ 6 months
- Experiment started in November 2025

Stay tuned...

